MILITARY SPECIFICATION

MANUALS, TECHNICAL: OPERATION AND MAINTENANCE INTRUCTIONS FOR OPERATORS OF VARIOUS TYPES OF EQUIPMENT; PREPARATION OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains the detail requirements for the preparation of technical manuals covering operation and maintenance instructions for operators of the following types of equipment:

(a) Hand and portable weapons
(b) Combat vehicles (LVT's, tanks, etc.)
(c) Trucks
(d) Generator sets
(e) Construction and materials-handling equipment
(f) Radios
(g) Test and diagnostic equipment (optional)
(h) Audio visual equipment

These operator's manuals will be identified by a "-10" following the number assigned to the document.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation-packaging, Methods of.
MIL-L-22690	- Lubrication Charts, Plates, and Manuals; Preparation and Production of
MIL-M-0038784	- Manuals, Technical: General Requirements.
MIL-M-63001	- Manual, Technical, Basic Isswg Items List, Items Troop Installed or Authorized List, Repair Parts and Special Tools List.

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STANDARDS

MILITARY

MIL-STD-12	- Abbreviations for Use on Drawings, Specifications, Standards, and in Technical Documents.
MIL-STD-129	- Marking for Shipment and Storage.

PUBLICATIONS

U.S. MARINE CORPS

TM-4700-15/1	- Marine Corps Tactical Equipment Records.

DEPARTMENT OF DEFENSE

DOD 5200.1-R	- Information Security Program Regulations.

(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting office.)

3. REQUIREMENTS

3.1 General. These manuals shall describe in accurate and appropriate detail the operation and maintenance authorized for the operator/crew. When specified (see 6.2), the procuring activity shall furnish an outline which completely reflects the organization, style, format, illustrative techniques and extent of coverage. During preparation of the manual, the following guidelines shall be used:

(a) Accuracy of technical content and ease of use are the primary requirements of all material included in the manual.

(b) Sufficient information shall be included so that a minimally qualified operator can operate the equipment safely and efficiently. Coverage shall be governed by the fact that the manual shall be suitable for use by personnel of the least possible experience who could reasonably be expected to operate the equipment. Consideration shall be given to the fact the more complex the equipment, the more experienced the crew will have to be that is going to operate it. When necessary, the contractor should request the procuring activity to assist in determining the content and the level of manual coverage based on the experience level of the assigned crew and the mission.

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(c) The manual shall not be used to teach basic duties, but how to apply knowledge of basic duties to the operation of the system. Emphasis shall be placed on any variations to basic duties which may be created by the unique characteristics of the system.

(d) Emphasis shall be placed on designing a simplified manual which contains methods and techniques which effectively present the required information to the user. Examples are included throughout this specification. During preparation, special consideration should be directed to the following areas:

1. Integrate text and illustrations into modules of information. Do not separate text from illustrations when the text can effectively be integrated with the illustration.

2. Use columnar or tabular format for procedures which cannot be satisfactorily illustrated.

3. Paragraph numbering is optional for the text depending upon the particular style of presentation selected. Referencing, both internal and external, shall be kept to a minimum.

4. Use animated drawings/cartoons in the manual when the style of the manual lends itself to this type of presentation.

5. Narrative text (those pages that consist of not less than 200 words in consecutive sentences per page) shall conform to the following readability standards:

Standard No. 1. The average sentence length shall not exceed 200 words. The desired length is 1 to 17 words.

Standard No. 2. The average word length shall not exceed 1.60 syllables. The desired length is 1 to 1.50 syllables. The length of a word is measured by the number of its syllables. Mandatory words shall be excluded from this restriction.

Standard No. 3. The percent of personal sentences shall not be less than 15 percent of the total number of sentences in the narrative text. Twenty percent and above is desired.

6. Make maximum use of the various illustrative techniques, i.e., functionals, cut-away views, phantom views, locator views, pictorial diagrams, and specially designed pictorial techniques to portray operating and maintenance procedures.

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7. Make maximum use of line drawings rather than half tone illustrations to insure clarity.

8. Highlight important information needed by an experienced, high skill user to enable him to utilize essential information quickly after he is thoroughly familiar with the detailed procedure. This allows the high skill user to quickly scan the procedure and pick out the information he needs without reading the entire procedure. Underlining, boldface type, italics and boxing may be used to highlight information.

3.2 Size. Size of the manual will be specified by the procuring activity (see 6.2) in accordance with MIL-M-0038784, except that size 6-1/2 inch x 9-1/2 inch is also authorized.

3.3 Skill enhancement materials. Skill enhancement materials shall be used throughout the manual to increase the operator's understanding of the equipment. The operating and maintenance procedures in a manual cannot possibly cover all situations the manual user may encounter (especially in combat situations), therefore, this additional understanding of his equipment will motivate him and help him become more effective in a larger variety of situations. Skill enhancement materials include:

(a)	Overviews	- For skill reinforcement of complicated tasks or information.
(b)	Summaries	
(c)	Procedure descriptions	- For skill enhancement beyond present job.

3.3.1 Overviews and summaries. Skill enhancement materials shall be provided for complicated tasks or information which cannot be easily learned after a few times of use. The skill enhancement material shall be in the form of overviews and summaries located before and after the complex task or information. Where the material to be reinforced is lengthy, the overview and summary information shall be interspersed throughout the material. The overview presents the material which will be covered and prepares the operator to use it. The overview describes the material which is to be memorized and contains the objectives. Summaries will review the important information, stressing the material which was to be memorized, and providing a means for the user to check his understanding of the material and the ability to perform his job to normal standards. Part of the review should include questions to be answered by the user. A typical overview is shown in Figure 1. A typical summary with questions is shown in Figure 2.

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Overview of Operating Instructions for the M97A1 Weapon

This chapter has some very important information that you need to know about your weapon. After reading and studying this chapter, you should be able to:

a. Locate and use all parts on your weapon.

b. Field strip and clean your weapon.

c. Fix misfires.

d. Make sure you won't be hurt by a possible cookoff.

e. Use the sling.

f. Adjust your sights.

Figure 1. Overview

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Summary of Operating Procedures for M97A1 Weapon

In this chapter you should have learned how to:

a. Locate and use all parts of your weapon including the charging handle, forward assist, slip ring, bolt cam pin, bolt catch, ejection part cover, flash suppressor, front sight, magazine release cover, and selector lever (pages 3-36).

b. Perform the basic 19 steps of field stripping and cleaning (pages 18-21).

c. Assemble the firing pin, oil the firing-pin hole, recognize defective ammo, and remove carbon on the firing pin to prevent misfires (pages 16-18).

d. Avoid cookoff after a misfire by removing the cartridge and the magazine within 10 seconds, or if cartridge is stuck, by waiting 15 minutes (page 15).

e. Adjust the sling for steadying the weapon while firing (page 5).

f. Zero in your sights at 250 meters (pages 10-11).

You should know the answers to these questions.

a. Do you attach the bayonet to the front sight or the flash suppressor (page 4)?

b. In field stripping your rifle, which do you remove first, the charging handle or the bolt carrier and bolt (page 19)?

c. How can you tell a blank cartridge in the dark (page 16)?

d. If your rifle won't fire, what should you do first (page 14)?

e. If you're zeroing in your sights and you hit the target too far to the right, how would you adjust the sights (page 10)?

ANSWERS:

a. Front sight. b. Remove the bolt carrier and bolt. c. A blank has a flat tip. d. Tap upward on the magazine. e. Move the rear sight windage drum counterclockwise (the opposite way a clock moves).

Figure 2. Summary with Questions

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3.3.2 Procedure descriptions. The best time for the operator to improve his understanding of the workings of the equipment is while carrying out the procedures in the manual. Skill enhancement materials shall be included in procedures for the following purposes:

(a) Provide the user with an opportunity to better understand his job.
(b) Motivate the user to do a better job.
(c) Enable the user to gain general technical knowledge which he can transfer to a similar job or other equipment.
(d) Enable the user to progress to the point where he thinks for himself.

Skill enhancement materials shall be used only in cases where obvious skill upgrading benefits will accrue - not as filler material with every procedure or step. Skill enhancement materials shall include the following information:

(a) Purpose. This shall be a brief statement describing what the task will accomplish.
(b) General technical principles. This shall be a brief functional description of the general technical principles related to the task. This shall be written in general terms so that the user will understand how this equipment and similar equipment function. This part is included to increase the user's knowledge of the general type equipment he is using and enable him to transfer his specific equipment knowledge to similar equipment. An illustration or functional diagram may be included if it will aid in the explanation.
(c) Task description. This shall be a brief description of how the general technical principles relate to the specific task to be accomplished. This also gives the operator an overview of the procedure by explaining the purpose of each major group of steps that form the procedure. An illustration or functional diagram may be included if it will aid in the explanation of the task.
(d) Remarks column. The remarks column in each procedure provides additional specific information about the task. Alongside each step, or group of steps, the remarks column explains what each action accomplishes. By following this description as he works, the operator will understand how the performance of each step accomplishes the purpose and objectives of the procedure.

See Figure 3 for a typical example of a, b, and c above.

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CHECKING TRACK TENSION

Checks track tension and makes any necessary adjustments.

It is very important to your safety that track tension is correct.

Tension Too High

Tank suspension will be too stiff so steering and performance will be affected.

Having track tension too high is like having overinflated tires on your car.

Tension Too Low

A track could be thrown off.

Having track tension too low is like having underinflated tires on your car.

Job Description

You check track tension by checking clearance between track and upper support wheels. A 6" x 6" x 1" wooden block is used like a big feeler gauge.

You can increase or decrease track tension using your wrench on the adjusting link assembly. Turning this assembly changes the distance between support wheels.

Figure 3. Procedure Description

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3.4 Typical arrangement of the manual. The manual shall contain the information listed below, as applicable:

Front Matter	
How-To-Use This Manual	
Chapter 1	- Introduction
Chapter 2	- Operating Instructions
Chapter 3	- Maintenance Instructions
Chapter 4	- Maintenance of Auxiliary Equipment
Chapter 5	- Ammunition
Appendix A	- References
Appendix B	- Basic Issue Items List and Items Troop Installed or Authorized List
Reference Information	
Index	

Generally, each item specified shall be included in the manual, unless the item is not applicable to the particular item of equipment or a deviation has been granted by the procuring activity. (Applicable pages, chapters, sections and paragraphs shall be numbered in sequence; the term "Not Applicable" shall not be used). Further, additional information and instructions which are needed for a specific item of equipment, but are not specified herein, shall be recommended to the procuring activity for inclusion in the manuals for that item of equipment.

3.4.1 Front matter. The front matter shall contain the following elements as applicable:

3.4.1.1 Cover. A typical cover for a standard size (8-1/4 inch x 10-3/4 inch) manual is shown in Figure 4. A typical cover for a pocket size (4 inch x 5-3/4 inch) manual is shown in Figure 5. Besides the usual general front cover information, these covers have two important additional features. The first feature is the front cover table of contents which shall include only those entries the operator will use most often. The second feature is a picture on the front cover showing the equipment covered by the manual. A warning summary shall appear on the inside front cover to provide the operator with the important warnings that apply throughout the manual.

3.4.1.2 Warning and first aid data page. The warning and first aid data page shall be included only if warnings and first aid data cannot be effectively summarized on the inside front cover. When included, the warning and first aid data page shall provide the operator with a summary of the important warnings which apply throughout the manual. The page shall also include any first aid data which the operator was not taught in basic training but may be needed due to the dangers which can be

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OPERATOR'S MANUAL

TM-09286-10

GUN, ANTIAIRCRAFT ARTILLERY
TOWED: 20 MM, XM167
(1005-999-4391)

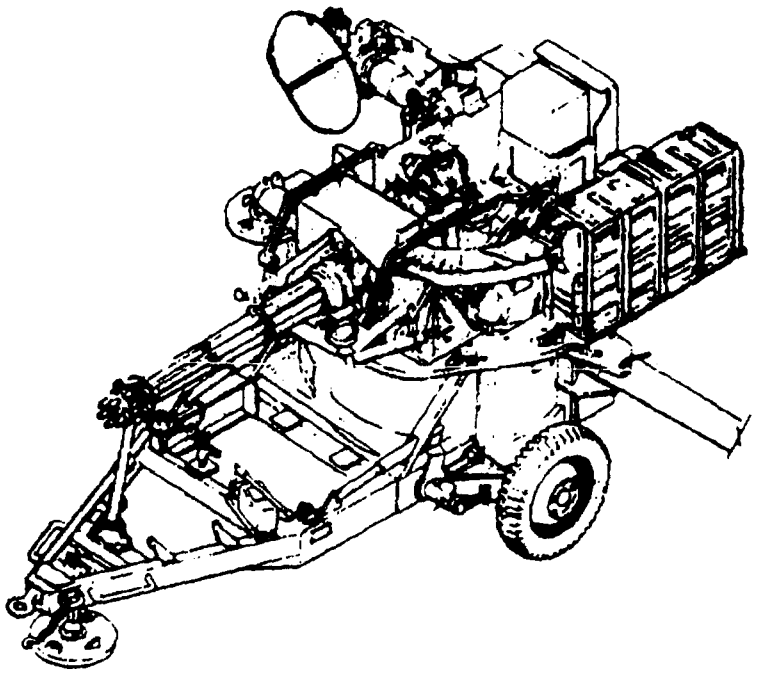


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HEADQUARTERS, U. S. MARINE CORPS

December 1973

Figure 4. Cover for Standard Size (8-1/4" x 10-3/4") Manual

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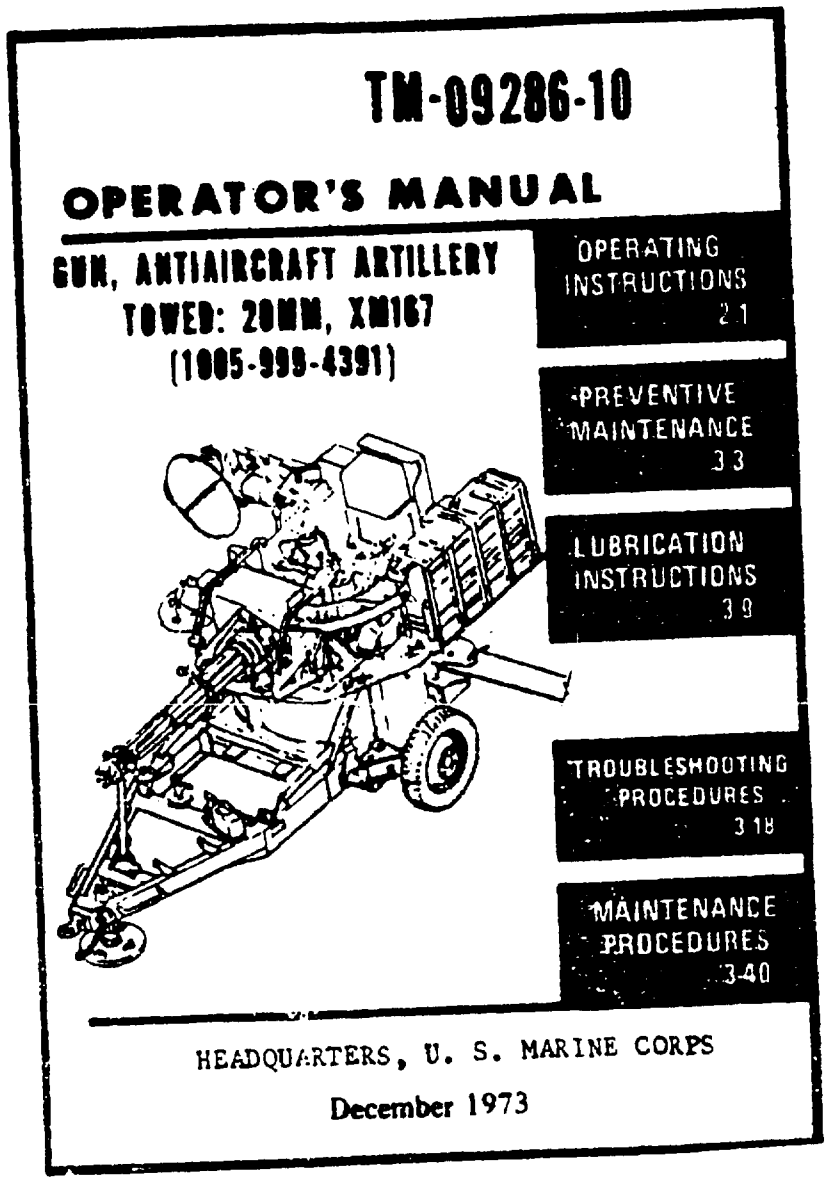


Figure 5. Cover for Pocket Size (4" x 5-3/4") Manual

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encountered with the equipment. A typical warning and first aid data page is shown in Figure 6.

3.4.1.3 Table of contents. A typical table of contents is shown in Figure 7. An entry which is a duplicate of an entry on the front cover shall be enclosed in a box. This duplication of information is a precautionary measure because the front cover may be torn off due to excessive or abusive use. When authorized by the procuring activity, attention directing features may be used to improve access to information in the manual - such as tabs, colored pages, cut-away edges, bleeding, etc.

3.4.2 How-To-Use This Manual information. "How-To-Use This Manual" information shall be located before the first chapter in manuals which are designed for use by the inexperienced, novice operator. This section shall include the following parts:

(a) User prerequisites
(b) Manual overview
(c) Practice problems

3.4.2.1 User prerequisites. The prerequisites listing shall be based upon the assumptions about the operator made by the writer during preparation of the manual. The purpose of the prerequisites is to enable the operator to assure himself that he is qualified to use the manual to perform his job, or if not, to determine exactly what background he lacks and where to get it. The listing shall include the general assumptions made about all users of the manual (both experienced and inexperienced) and the specific differences in assumptions made about experienced and inexperienced operators. For example, that the experienced operator knows the location of all controls and indicators but the inexperienced operator does not. A typical user prerequisites listing could be:

(a) MOS 64B
(b) A valid driver's license
(c) Attended the nontracked vehicle school
(d) At least 2 months of practice driving nontracked vehicles

3.4.2.2 Manual overview. This part shall clearly and concisely explain important features of the organization and contents of the manual, stressing any of the unusual or unique features. Any special or unusual features shall be described in detail. This part may be combined with practice problems in 3.4.2.3 where the combination effectively covers the features. The preferable style of description of any special or unique feature of the manual is a short, interesting, real-life example which makes use of the feature to be described, and enables the user to practice using it as he reads the example.

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WARNING

COOKOFFS AND ACCIDENTAL FIRINGS

- Always clear cannon before using, servicing, or maintaining.
- Make sure weapon is trained on the target and all personnel are clear of the barrels.
- When gun has been in action, clear cannon before anyone moves in front of muzzle.
- Because of possible cookoff, never attempt to remove a round that is chambered in a hot weapon.
- Don't rechamber any cartridge - it can easily explode prematurely.

EXPLOSIVE FUEL

- Provide metallic contact between container and generator to eliminate static electricity, which can cause explosion.
- Turn off all electrical switches before refueling generator.

CAUSTIC CHEMICALS IN BATTERIES

- Use rubber gloves and apron to avoid severe burns.
- If chemicals get on your skin, clothes, or equipment, wash immediately with water.
- If chemicals get in your eyes, wash them with plenty of water and get medical help immediately.

Figure 6. Warning and First Aid Data Page

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TM-09286-10

Operator's Manual

CARRIER, COMMAND AND RECONNAISSANCE: ARMORED, M114/M114A1E1

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Figure 7. Table of Contents

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3.4.2.3 Practice problems. This part shall provide one or more practice problems. The problems shall be written so that the result of using the problems is that the operator has been led through most of the manual and its indexes, stressing any new or unique features of the manual. The problems shall be real world problems which the operator could encounter while on the job. A typical practice problem is shown in Figure 8.

3.4.3 Introduction chapter. The purpose of this chapter is two-fold. The first purpose is to provide the operator with standard data required in all manuals (i.e., forms and record data). The second purpose is to familiarize the operator with the equipment. This familiarization is provided through both a physical and functional description of those major equipment components the operator is likely to interface with. This chapter shall consist of the following sections:

Section I	- General information
Section II	- Equipment description
Section III	- Technical principles of operation

3.4.3.1 Section I - General information. For operator orientation purposes, this section shall begin with an illustration which provides a full external view of the equipment. A typical full view illustration is shown in Figure 9. This section shall also consist of the following information:

3.4.3.1.1 Scope. The scope paragraph shall contain a brief statement regarding the following topics, as applicable:

(a) Type of manual
(b) Model number(s) and equipment name(s)
(c) Purpose of equipment
(d) Special limitations on equipment
(e) Type of personnel
(f) Special inclusions in manual

A typical scope paragraph for a multiple-configuration equipment is shown in Figure 10.

3.4.3.1.2 Maintenance forms and records. The manual shall include samples of the forms most often used by the operator (see Figure 11). In addition, this paragraph shall contain the following statement:

"Marine Corps forms and procedures used for equipment maintenance will be those prescribed by TM-4700-15/1."

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PROBLEM

You have been on maneuvers for the last 18 days in the worst weather you have ever seen. But at 1500 hours today you will be driving back to home base and much needed weekend leave. At 1400 hours you check out your truck and find that the engine won't start. During training they told you the surest and quickest way to fix your truck was by using your manual, so you grab your manual.

1. How do you start?

   Turn to the cover of your manual.

   On the right side you will find a listing for "TROUBLESHOOTING Symptom Index." It tells you to go to page 3-105. (To find that section quickly, open the manual by using the black tab on the side of the manual that lines up with the "TROUBLESHOOTING Symptom Index" listing.

2. What kind of problem do you have?

   Open your manual to page 3-105.

   In the "Symptom Index" find the listing for the equipment category, "ENGINE SYSTEM." There you find your symptom which is "Engine cranks but doesn't start." Next to the symptom is the page number of the troubleshooting procedure that will help you solve your problem. This procedure is on page 3-122.

3. How do you determine what is causing your problem?

   Turn to page 3-122.

   You find there the troubleshooting procedure you need. The procedure has columns with headings: "Step," "Action," "Indication," "Remarks." Starting at step 1, read the procedure horizontally across the page. Each step tells you what to do ("Action" column) and what to look for ("Indication" column) at that time. The "Remarks" column has answers or explanations to some of the questions you may have in your mind as you are following the instructions.

   Let's assume when you check the air filter (step 6) the engine starts. Now you know you have a clogged air filter. To replace the filter element, step 6 tells you to go to the maintenance procedure on page 3-142.

4. How do you fix it?

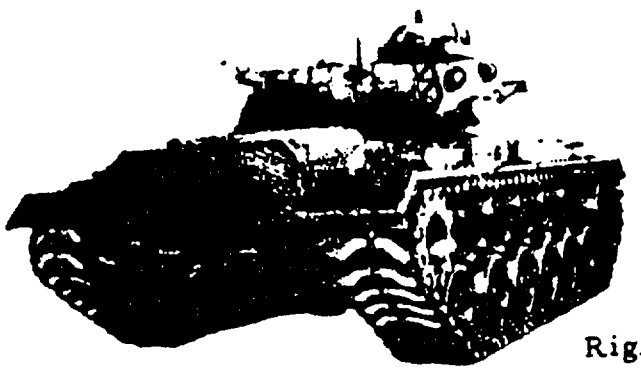
   Turn to page 3-142.

   This is the maintenance procedure for replacing the air filter element. This procedure is arranged the same way as the troubleshooting procedure. Start at step 1, read horizontally across, and perform the action in each step. When you finished the last step, you will have replaced the air filter element. Now you are ready to go home.

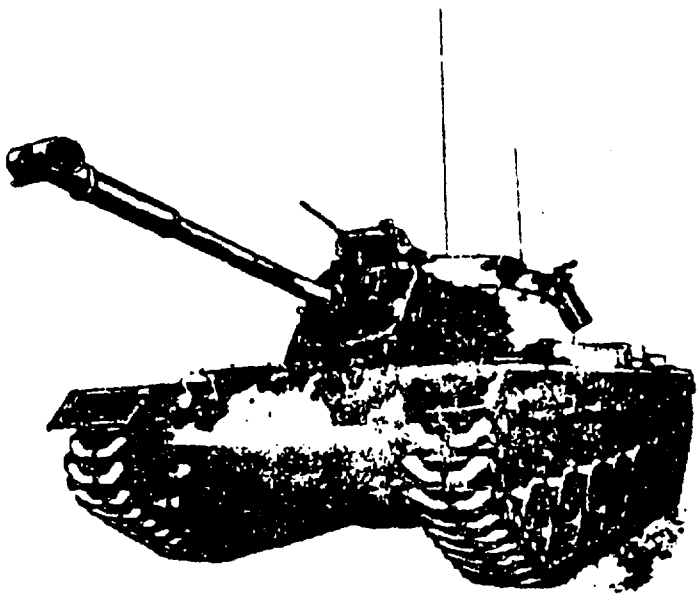
Figure 8. How-To-Use This Manual Practice Problem

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90 MM GUN
FULL TRACKED, COMBAT TANK M48A3



Right-rear



Left-front

1-1

Figure 9. Full View Illustration

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SCOPE

Use this manual for operating and maintaining the following carriers:

M113A1	Full-tracked, armored personnel carrier
M577A1	Light-tracked, command post carrier
M106A1	Self-propelled, 107 mm mortar system
M132A1	Self-propelled, flame thrower
XM741	Self-propelled, 20 mm anti-aircraft artillery gun chassis

Your manual:

1. Contains information for safe and efficient operation of your equipment. These instructions provide you with a general knowledge of the equipment, its characteristics, and usual and unusual operating procedures.

2. Provides you with troubleshooting procedures and authorized repair procedures so you can keep your equipment properly operating.

3. Provides best possible operating instructions under most circumstances. Multiple emergencies, adverse weather, terrain, etc. may require modification of the procedures.

4. Takes a "positive approach" and normally states only what you can do. Unusual operations or configurations are prohibited unless specifically included. Clearance must be obtained from your supervisor before any questionable operation, which is not specifically permitted in this manual, is attempted.

Figure 10. Scope Paragraph

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DATE	VEHICLE OPER. (HOURS) (MILES)	GALLONS GAS. ADDED	POL USED (QUARTS) (GALS) TYPE	TYPE	TYPE	TYPE	TYPE PREVENTIVE MAINTENANCE	REMARKS/MALFUNCTIONS	VEHICLE WAS OPER.	VEHICLE WAS INOP.	CREW CHIEF SIGNATURE
1	2	3	4	5	6	7	8	9	10	11	12

TRACKED VEHICLE DAILY LOG (4700)
NAVMC 10393 (REV. 8-80)

VEHICLE SERIAL

(Continue on Reverse)

Figure 11. Sample of Applicable Form

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3.4.3.1.3 Reporting errors and improvements. Instructions for reporting errors in, and recommendations for, the text shall be included in each manual in the style of the publication.

Example:

Every effort is made to keep your manual current. Review conferences with operating personnel and a constant review of accident and test reports assure the latest data in your manual.

We cannot correct an error unless we know of its existence. In this regard, it is essential that you do your part. Comments, corrections, and questions regarding this manual or any phase of the manual program are welcomed. Reports should be submitted on the postpaid postcard inserted in this manual.

Extensive or additional reports should be submitted on NAVMC 10772 and forwarded direct to the Commandant of the Marine Corps (Code LMO-1), Washington, D. C. 20380.

See Figure 12 for samples of tear-out cards, each of which includes a brief questionnaire for manuals described herein.

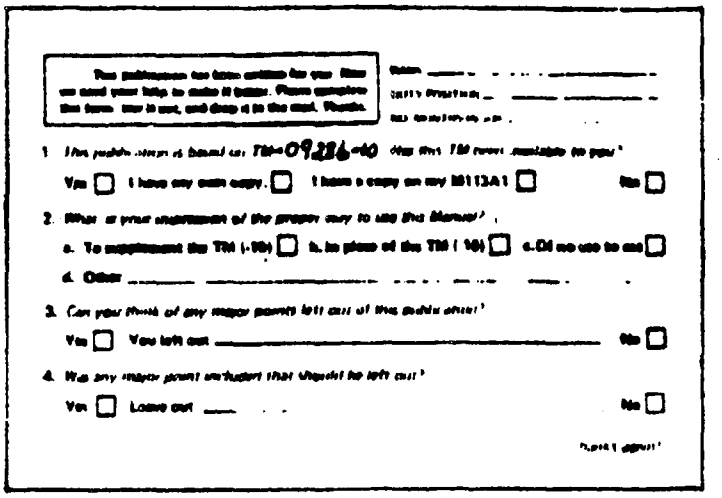
3.4.3.1.4 Nomenclature cross-reference. When nomenclature used in the manual differs from official nomenclature, an appropriate cross-reference listing shall be included. See Figure 13 for example.

3.4.3.1.5 List of abbreviations. Special or unique abbreviations, acronyms, and symbols used within the manual which are not defined in MIL-STD-12 shall be listed in the manual. See Figure 13 for example.

3.4.3.2 Section II - Equipment description. This section shall consist of the following information:

3.4.3.2.1 Equipment purpose, capabilities and features. This information shall consist of the purpose, general capabilities, applications, special features, and limitations of the overall equipment, emphasizing the data which is helpful in the operation of the equipment. The equipment type shall be noted as well as its portability or mobility operational and special environmental features, and remote control features. Components and their functions shall not be described unless essential to continuity, as this information is covered in following paragraphs. When the equipment covered constitutes a system which may vary in scope and application or which may have several applications within a system, a brief explanation of the multiple usage shall be made and a simple diagram showing the overall aspects of a typical application shall be included. This material shall refer to the system planning section in the installation chapter for information and detailed techni-

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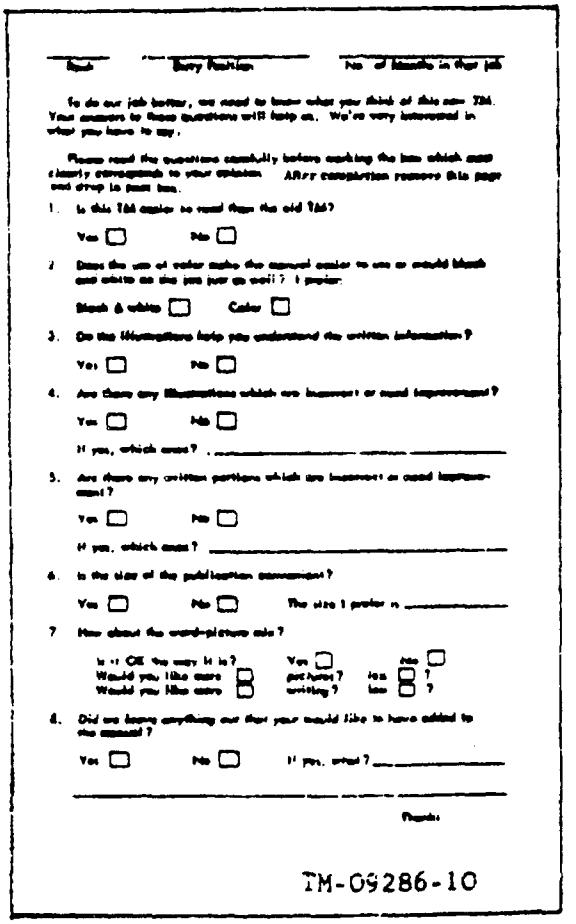


Figure 12. Sample User Feedback Postcard (Back side shown) (Front side contains applicable address)

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REFERENCE INFORMATION

This listing includes the nomenclature cross reference list, list of abbreviations, and explanation of terms (glossary) used in this manual.

A. NOMENCLATURE CROSS REFERENCE LIST

Common Name	Official Nomenclature
Loader-Transporter (LT)	Launcher, Guided Missile, Carrier Mounted:M752
Handling Unit	Loader-Transporter, Guided Missile: M688
Launcher (LZL)	Handling Unit, Guided Missile Equipment:M39
Mobility Kit	Launcher, Zero Length, Guided Missile: M740

B. LIST OF ABBREVIATIONS

ac	alternating current
amp	ampere
AR	Army regulation

C. GLOSSARY

Adjust	To bring within specified operating range
Inspect	To determine serviceability of an item by comparing its physical, mechanical and electrical characteristics with established standards
Replace	To replace unserviceable items with serviceable like items

Figure 13. Sample of Combined Reference Information

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cal considerations. A typical equipment purpose, capabilities and features paragraph is shown in Figure 14.

3.4.3.2.2 Location and description of major components. External and internal views of the equipment shall be used to show general features, essential operating items, and all major components and accessories. The major items in the views will be explained by the use of keyed text. This material shall provide a physical description of the equipment for the operator. When the equipment is designed for use with other equipment, one or more diagrams shall be used to illustrate the normal use of the equipment. Only information pertaining to the operator shall be included. Information to be included as specification data shall not appear here. A typical example of location and description of major components is shown in Figures 15 and 16. For simple equipment, the location and description of major components information could be combined with the controls and indicators section (see 3.4.4.1) and not appear here.

3.4.3.2.3 Differences between models. Significant differences between equipment models, or units of the same model, which affect operator actions shall be described and the extent of interchangeability shall be indicated. These differences shall be related explicitly to equipment model, part number, or serial number ranges in such a manner that an operator can clearly identify his own equipment configuration. When model or unit differences exist, but have no effect on operation or maintenance actions, either throughout the manual or in a given portion, this fact should be so stated, and a common nomenclature should be established. A typical example of a differences between models table for multiconfiguration equipment is shown in Figure 17.

3.4.3.2.4 Specifications. Performance data including numerical and other specification-related data applying to the operator's function shall be provided. The specification data shall summarize the specific capabilities and limitations of the equipment and other critical data needed by the operator for operation and maintenance of the equipment. Such data as shown in Figures 18 and 19 should be included. See Figure 20 for a typical example.

3.4.3.3 Section III - Technical principles of operation. This section shall contain a functional description of equipment operation. Only information pertaining to the operator shall be included. The purpose of this section is to tell the operator how the controls and indicators he uses interface with the rest of the equipment. Hence, the emphasis of this section should be the functional description of the controls and indicators and a deemphasis of the highly technical principles of parts of the equipment the operator does not interface with. A typical example for part of the technical principles of operation for complex equipment is shown in Figure 21. For simple equipment, the technical principles of operation information could be combined with the controls and indicators section (see 3.4.4.1) and not appear here.

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Purpose of TOW Weapon System

A crew portable, heavy antitank weapon designed to attach and defeat armored vehicles and other targets such as field fortifications.

Capabilities and Features

Major weapon system components:

a. Launcher
   1. Launch tube
   2. Traversing unit
   3. Missile guidance set
   4. Battery assemblies (2)
   5. Optical sight
   6. Tripod

b. Missile (tube-launched, optically tracked, wire-command-link)

All weather operational

High first-round hit probability

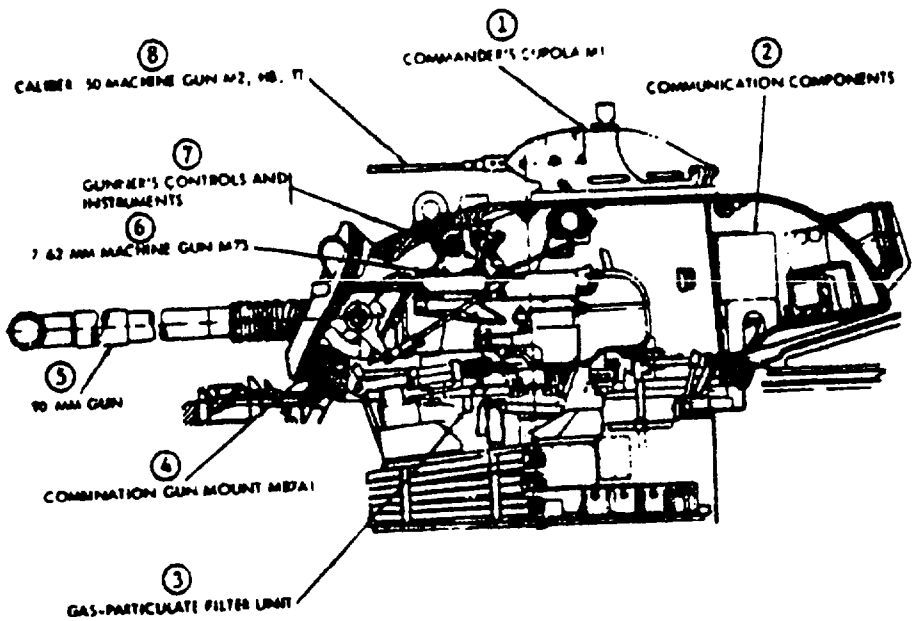
Highly portable

Built-in self-test circuits

Figure 14. Equipment Purpose, Capabilities, and Features

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LOCATION AND DESCRIPTION OF TURRET COMPONENTS



1-4

Figure 15. Location and Description of Major Components (Sheet 1 of 2)

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TURRET

Fits into hull opening. Made from one-piece homogeneous armor steeling casting. Provides openings to accomodate combination gun mount M87A1, commander's cupola M1, and other external turret components. Rotates 360° to enable 90 mm gun, 7.62 mm machine gun M73, and caliber machine gun M2, HB, TT to point at targets in any direction. Permits commander, gunner, and loader to move as guns move.

(1) COMMANDER'S CUPOLA M1. Provides self-contained enclosure for commander. Protects commander from direct or overhead bursts. Contains caliber .50 machine gun M2, HB, TT and necessary instruments and controls to lay machine gun in azimuth and elevation, and fire machine gun. Provides commander with external vision through vision blocks. Enables machine gun to be serviced and operated under cover.

(2) COMMUNICATION COMPONENTS. Encloses ground-to-ground communication system units.

(3) GAS PARTICULATE FILTER UNIT. Provides filtered air from 12 cfm filter through individual hoses to masks for commander, gunner, and loader.

(4) COMBINATION GUN MOUNT M87A1. Consists of concentric hydrospring recoil mechanism and integral mount, and rigidly attached gun shield. Pivots in elevation on turret trunnions. Supports 90 mm gun and 7.62 mm machine gun M73. Covers turret gun opening throughout elevation range by means of a cast steel shield.

(5) 90 MM GUN. Consists of quick-change tube retangular breeching, vertical-sliding breechlock, and counterrecoil-actuated breech mechanism with provisions for manual operation. Includes bore evacuator and muzzle blast defects on end of quick-change tube. Fires either electrically or mechanically.

(6) 7.62 MM MACHINE GUN M73. Light weight, air cooled, metallic link belt fed; has short receiver, is recoil operated (with gas assist to boost recoil), and is designed with quick-change barrel with fixed headspace and functioning from retracted position of barrel extension assembly.

(7) GUNNER'S CONTROLS AND INSTRUMENTS. Includes mechanical, electrical, and hydraulic gun control and sighting components for 90 mm gun and 7.62 mm machine gun M73.

(8) CALIBER .50 MACHINE GUN M2, HB, TT. Automatic, recoil-operated, link-belt fed, air-cooled, heavy barrel (HB), turret type (TT). Fires either manually, using solenoid trigger, or electrically by pushing electrical firing button.

1-5

Figure 16. Location and Description of Major Components (Sheet 2 of 2)

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CARRIER FUNCTION	M113A1	M577A1	M106A1	M125A1	M132A1	XM741
Personnel/cargo	X					
Command post		X				
Mortar carrier			X	X		
Flame thrower					X	
Anti-aircraft gun carrier						X
ARMAMENT AND FIRE-CONTROL						
M10-8 flame thrower					X	
107-mm mortar			X			
81-mm mortar				X		
Caliber .50 machine gun	X		X	X		
7.62-mm machine gun					X	
I.R. periscope M19	X	X	X	X	X	X
Periscope M17	X	X	X	X	X	X
AUXILIARY EQUIPMENT						
Exhaust extension			X	X		X
Air grille curtain			X			X
Covered extension			X			
Generator set/cover						X
Auxiliary power unit			X			
Electronic heater kit			X			
Personnel heater kit	X	X	X	X		
Engine coolant heater kit	X	X	X	X		
Driver's windshield kit	X	X	X	X		X
Litter kit	X					
CBR kit*	X	X	X	X	X	X

*When available.

Figure 17. Differences Between Models Table

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TYPICAL EQUIPMENT SPECIFICATION	TYPE OF EQUIPMENT						
	Hand and Portable Weapons	Combat Vehicles	Trucks	Construction and Material Handling Equipment	Generator Sets	Radios	Test and Designator Lasers and Aerial Vision Lasers
Caliber of weapons	x	x					
Capacities (ammunition, cooling, fuel, gear case, hydraulic, oil, stowage, etc)	x	x	x	x	x		
Center of gravity	x	x	x	x	x		
Clearances	x	x	x	x			
Detection characteristics (frequency, range rate, etc)	x	x					
Dimensions (emplacement, travel, overall, component, etc)	x	x	x	x	x	x	x
Engine characteristics (bore, cylinder, rpm, horsepower, etc)		x	x	x	x		
Field of fire	x	x					
Field of view		x					
Gear speeds		x	x	x	x		
Ground pressure	x	x	x	x			
Land performance (speeds, towed speeds, grades, turning radius, etc)	x	x	x	x			
Loads (maximum)	x	x	x	x			
Make, model, equipment type, and major components	x	x	x	x	x	x	x
Muzzle velocity, pressure	x	x					
Operating modes	x	x	x	x	x	x	x
Operating temperatures	x	x	x	x	x	x	x
Optical characteristics	x	x					
Range of fire	x	x					
Rates of fire	x	x					
Rifling	x	x					
Slew rates	x	x					
Tire pressure	x	x	x	x			
Water performance	x	x	x	x			
Weights (components, overall, bridge classification, etc)	x	x	x	x	x	x	x
Accuracy (tolerances)	x	x	x	x	x	x	x
Antenna characteristics						x	x
Bandwidths						x	x
Baud rate						x	x
Calibration requirements						x	x

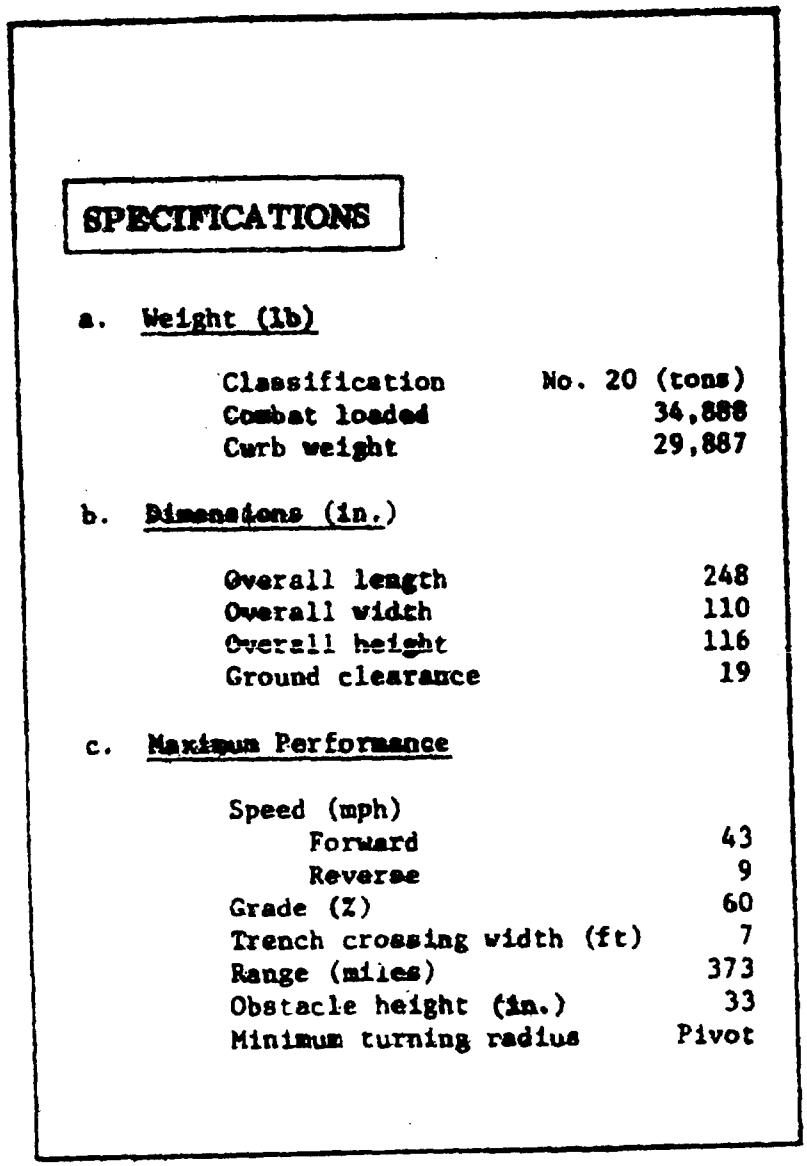
Figure 18. Specifications Required by Equipment Type (Sheet 1 of 2)

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TYPICAL EQUIPMENT SPECIFICATION	TYPE OF EQUIPMENT						
	Hand and Portable Weapons	Combat Weapons	Trucks	Construction and Material Handling Equipment	Generator Sets	Radios	Test and Diagnostic Equip and Audio Visual Equip
Channels (number, frequency) Cycles (duty, life, operating, etc) Frequency (operational, stability, bands, input, output, etc)					x x	x x x	x x x
Fuses Impedance (input, output, loop, etc) Meter sensitivity, ranges					x x x	x x x	x x x
Modulation characteristics Number of lines, links, trunklines Power (input, output, phase, drain, stability, etc) Radioactive material (type, quantity, etc)	x x	x x	x	x	x	x x x x	x x x
Receiver characteristics (sensitivity, gain, response time, tuning, range, etc) Signal sensitivity Term stability Transmitter characteristics (frequency, operating modes, power input, power output, range, signal strength, etc)						x x x	x x x
Voltage (input, output, stability, accuracy, source, dc, ac, frequency, range, etc)	x	x	x	x	x	x	x

Figure 19. Specifications Required by Equipment Type (Sheet 2 of 2)

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SPECIFICATIONS

a. Weight (lb)

Classification	No. 20 (tons)
Combat loaded	34,888
Curb weight	29,887

b. Dimensions (in.)

Overall length	248
Overall width	110
Overall height	116
Ground clearance	19

c. Maximum Performance

Speed (mph)	
Forward	43
Reverse	9
Grade (%)	60
Trench crossing width (ft)	7
Range (miles)	373
Obstacle height (in.)	33
Minimum turning radius	Pivot

Figure 20. Equipment Specifications

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BILGE PUMP SYSTEM

Removes accumulated water from carrier. Avoids flooding during adverse weather conditions or water operation.

STRAINER. Prevents debris from entering pumps by means of perforated metal screen to protect pump intakes.

FRONT AND REAR BILGE PUMPS. Remove water from vehicle through discharge tube by means of electrically operated rotor-type pumps. Activated by single switch on instrument panel.

LEFT FRONT AND RIGHT REAR BILGE OUTLETS. Throw off water.

SIGHT GLASS. Provides visual indication of water flow to enable driver to observe action of front pump.

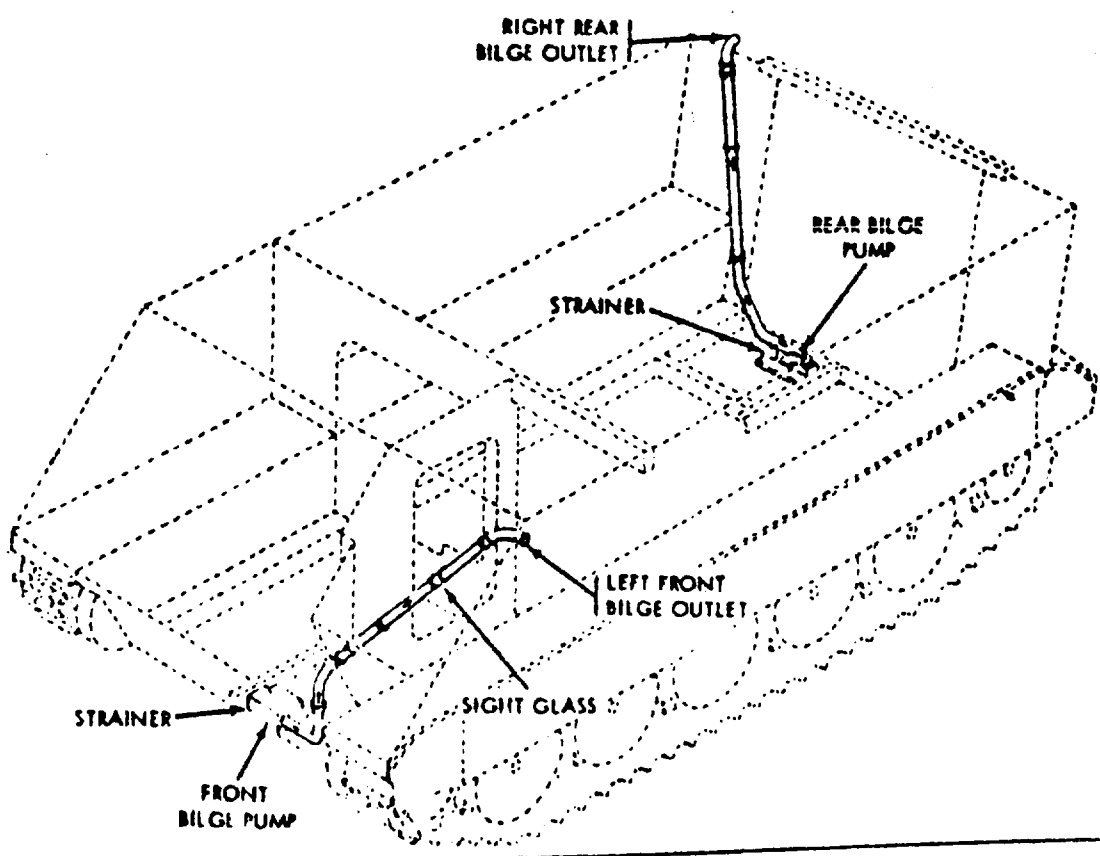


Figure 21. Technical Principles of Operation

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3.4.4 Operating instructions chapter. The objective of this chapter is to present those instructions needed so that the operator can effectively and efficiently use this equipment. Additional information is added only when that information will increase the operator's effectiveness. The level of detail must be appropriate for the user. Since it is important that the operator trust the instructions to prevent injuring himself or damaging the equipment, every effort shall be made to insure the instructions are completely accurate. This chapter shall consist of the following sections:

Section I	- Description and use of operator's controls and indicators
Section II	- Operation under usual conditions
Section III	- Operation under unusual conditions

This chapter shall begin with an alphabetical index of the chapter contents if reference to this shapter is included in the front cover table of contents (see 3.4.1.3).

3.4.4.1 Section I - Description and use of operator's controls and indicators. Illustrations showing nonplacarded operator controls and indicators shall be included. Each control and indicator, both placarded and nonplacarded, will be functionally explained from the point of view of the operator as part of the illustration. Names of items shall correspond to panel markings unless the marking is obviously unsuitable (such as a lengthy series of numbered positions). Items with an unsuitable panel marking, or no panel marking, shall be given an appropriate functional name which is used consistently throughout the manual. The function of each item shall be explained in terms of audible or visual indications, or in terms which do not describe internal circuit functions. A typical example of part of the description and use of the operator's controls and indicators for complex equipment is shown in Figure 22. A typical example for simple equipment is shown in Figure 23. Figure 23 is an example of where information from the "Location and description of major components" (see 3.4.3.2.2) and "Technical principles of operation" (see 3.4.3.3) was combined with description and use of operator's controls and indicators. If the equipment has numerous controls and indicators, it may be desirable to use a tabular format as shown in Figure 24.

3.4.4.2 Section II - Operation under usual conditions. Step-by-step instructions shall be given for all actions necessary to operate the equipment. Any combination of control settings which will cause damage to the equipment or create a hazard to personnel shall be explained as part of the appropriate procedures. The damage which might result shall be briefly mentioned and appropriate cautions and warnings shall be included. Essential illustrations for these instructions shall be

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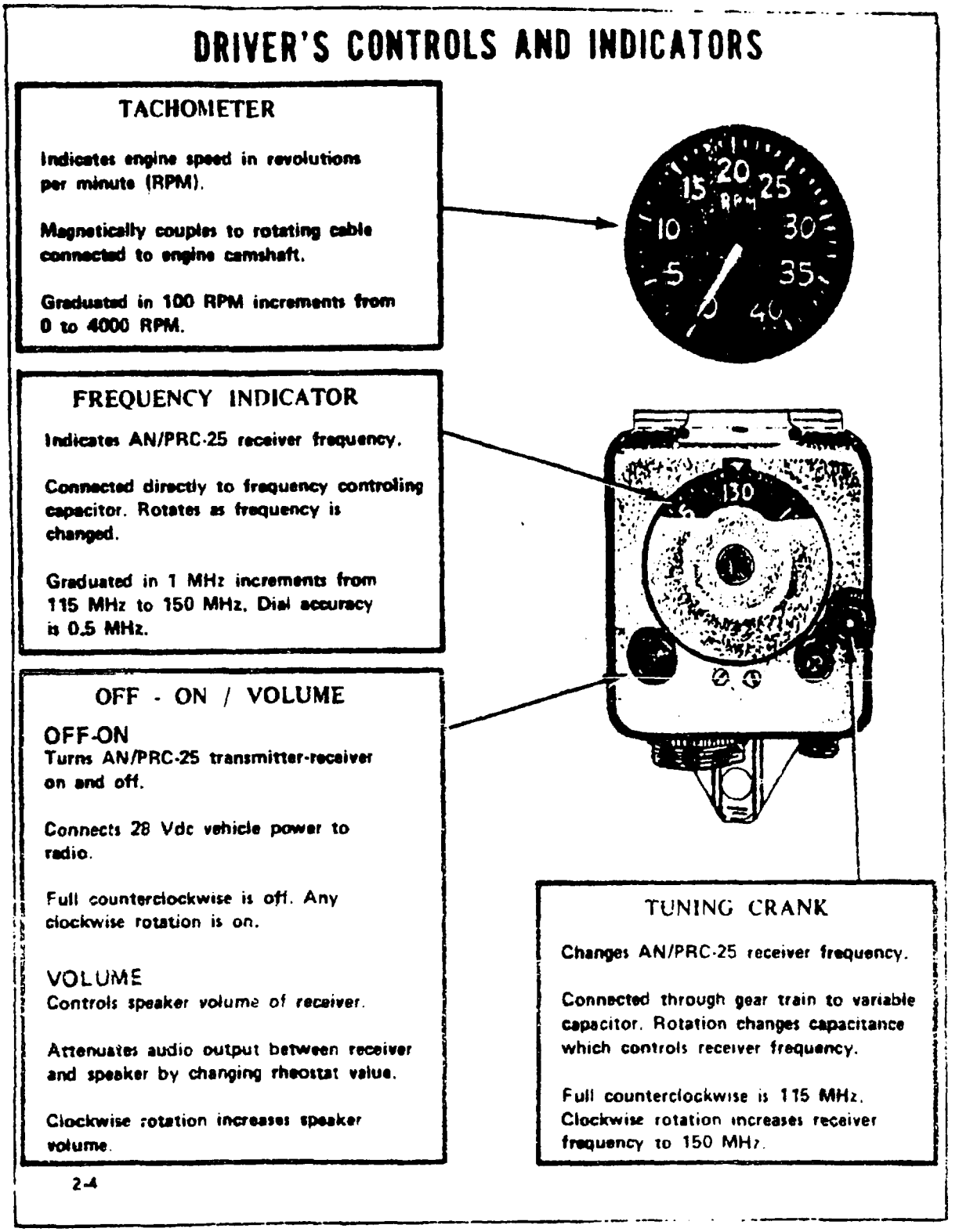


Figure 22. Description of Controls and Indicators

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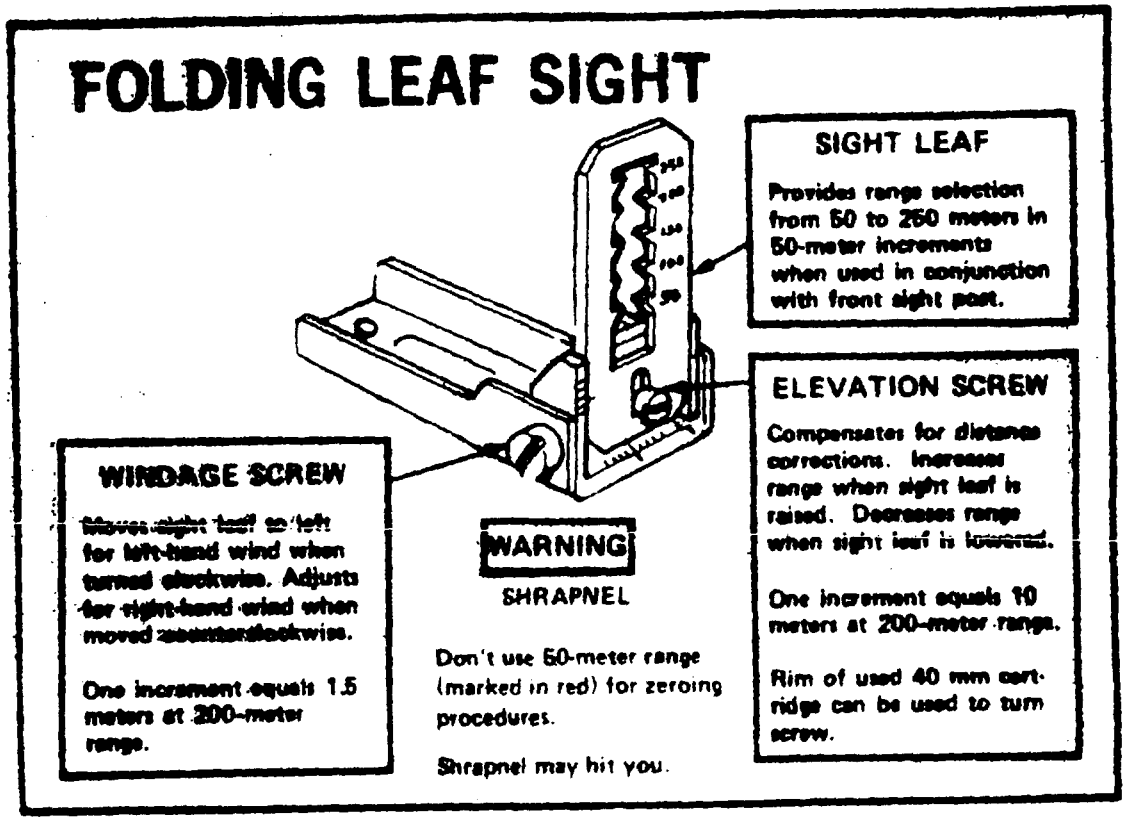
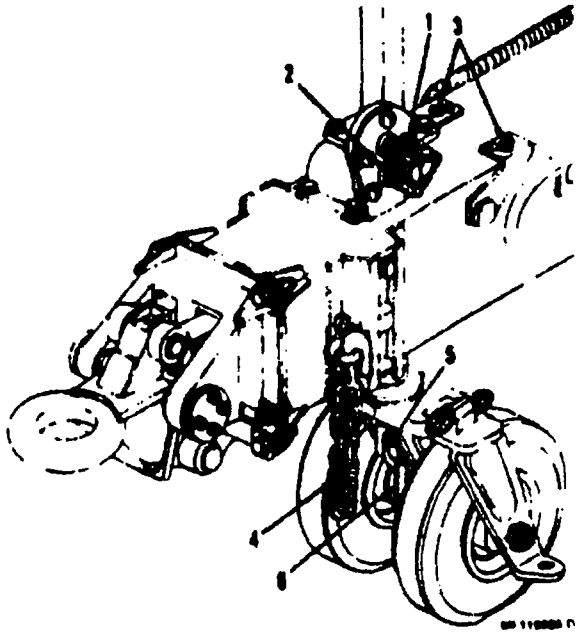


Figure 23. Description of Mechanical Controls

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Key	Control or Indicator	Type	Function
1	Cranking stud (High speed)		Extends or retracts caster jack.
2	Cranking stud (Low speed)		Extends or retracts caster jack and allows 2-man operation with missile round on board.
3	Caster up lock stud(2)		Engages with hole in lug on caster forks to secure caster wheels in stowed position.
4	Tow bar safety chain(2) shown in stowed position		Provides a safety factor when the tow bar is attached to a towing vehicle.
5	Recess for pry bar	3/4-in. dia. hole	Facilitates steering of caster wheels on difficult terrain.
6	Caster down lockpin	L-handle, push-pull pin retained by hairpin cotter	Locks caster wheels in down position.

Figure 24. Tabular Method for Description of Controls and Indicators

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included (see Figures 25 and 26 for typical operating procedures). The instructions shall cover the following:

3.4.4.2.1 Assembly and preparation for use. Necessary unpackaging, assembly, and installation instruction shall be provided. Instructions shall be given, as appropriate, for connecting to power and establishing the normal-, neutral-, zero-, center-, on-off-position of each switch, control or similar item when the equipment is in shutdown status.

3.4.4.2.2 Initial adjustments, daily checks, and self test. Instructions shall be given for any routine checks, self test, or adjustments which the operator should make prior to putting the equipment in operation.

3.4.4.2.3 Operating procedure. Operating instructions shall begin with the equipment in a shutdown status. Each mode of operation shall be covered individually, unless the procedures for the different modes are the same except for a few steps or for a specific segment. In the latter case, a single, integrated set of instructions shall be used giving alternate/additional steps or segments of the instructions for each mode of operation. Particular emphasis shall be given to cases where simultaneous operation of several controls is required. These operating procedures shall include the following:

(a) Procedure for equipment starting or turn on.
(b) Procedures for equipment operation. Instructions shall be given for equipment operation after equipment start-up for each of the applicable modes of operation.
(c) Procedure for placing equipment in standby condition. Instructions shall be given for placing the equipment in standby condition after operation. If different procedures are required for different modes of operation, instructions shall be given to cover each mode.
(d) Procedure for shutdown. Instructions shall be given for placing the equipment in shutdown status after the completion of operation in each of the applicable modes, including standby.

3.4.4.2.4 Operation of auxiliary equipment. This part shall contain instructions for the operation of auxiliary equipment supplied for use with the major item. Information shall be confined to that which is needed by the operator to properly identify, connect, operate, and protect auxiliary equipment while being used or transported with the major item. If no instructions on auxiliary materiel or equipment are needed, this part shall be omitted.

3.4.4.2.5 Preparation for movement. When the equipment is designed for routine movement (that is, it can be folded, telescoped, partially disassembled, or otherwise prepared for movement by the operator), step-by-step instructions for preparing the equipment for movement to the new location and for putting it back in service shall be included. Essential

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OPERATING SUSPENSION LOCKOUT

NOTE

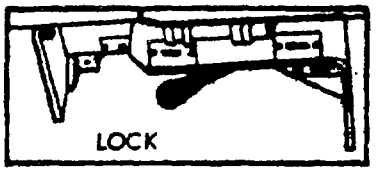
The master switch must be ON when you operate the suspension lockout.

The suspension lockout system lets you lock eight of the roadwheel arms in position to provide a stable platform for launching missiles. The lockout selector lever is on the front of the hydraulic power pack under the driver's seat.

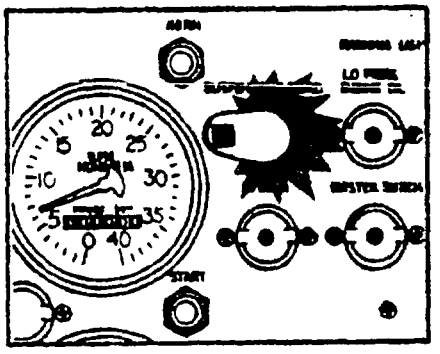
LOCKING OUT SUSPENSION

1 **If the range selector is NOT in N, an automatic interlock prevents the engine running above idle speed (650 - 700 rpm) when the suspension is locked out. This is to make sure the carrier won't start to move when you operate missile equipment.**

2 **Swing the lockout selector lever up to LOCK position.**

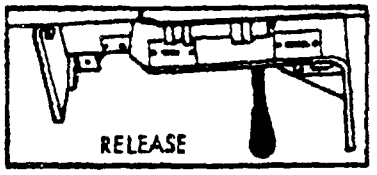


3 **Check the instrument panel to make sure the SUSPENSION LOCKED warning light comes on.**



RELEASING SUSPENSION LOCKOUT

1 **Swing the lockout selector lever down to RELEASE position.**



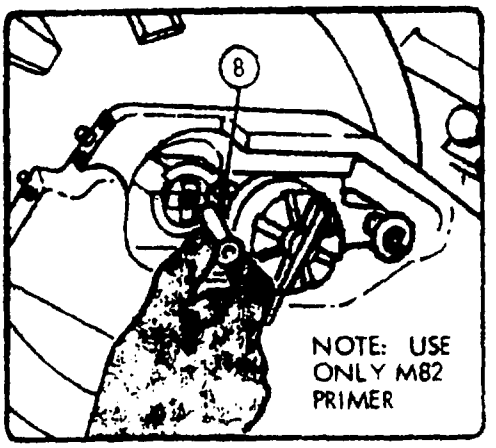
2 **Check the instrument panel to make sure the SUSPENSION LOCKED warning light goes out.**

3 **If the engine is running, make sure the hand throttle is in and the engine is idling at 650 - 700 rpm before you move the range selector out of N range.**

Figure 25. Operation Procedure Format (Example 1)

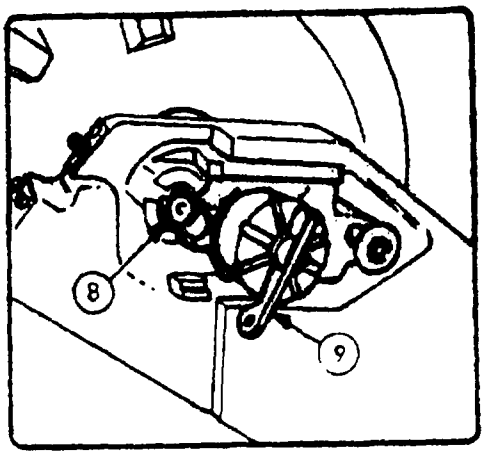
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LOADING AND FIRING 155MM CANNON - CONTINUED

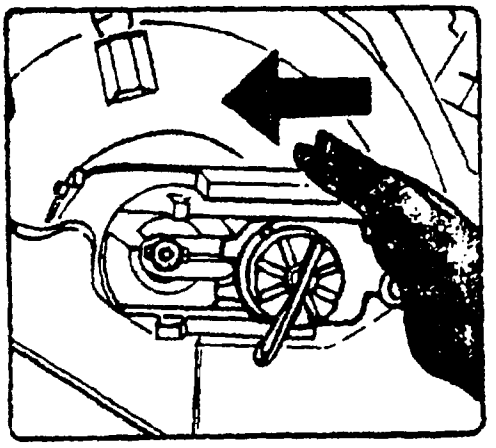


4 Insert primer (8) in primer chamber. Don't force it. If primer doesn't want to go in, your chamber is probably dirty. Clean both primer chamber and obturator vent hole (see items 37 and 40 in PMCS Table).

Note: Slide firing mechanism to prime position.

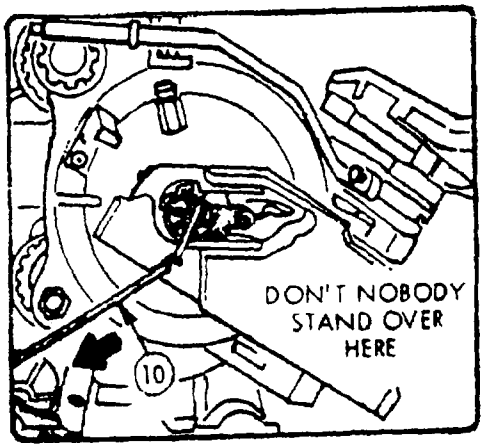


5 Make sure flanged head of primer (8) is firmly seated against extractor. Firing mechanism lever (9) can be in any of the slots on firing mechanism.



6 Slide firing mechanism block assembly to the left.

CAUTION: Be sure block assembly is all the way left. If it isn't you'll get a mushroomed firing pin which won't retract.



7 Attach lanyard (10) to eyelet on firing mechanism lever.

8 Stand out of recoil path and fire howitzer with a firm, steady pull on lanyard.

9 If howitzer fails to fire, refer to Section II. Precautions..... which follows on the next page.

Figure 26. Operation Procedure Format (Example 2)

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illustrations shall be provided, or referenced if suitable illustrations appear elsewhere in the same manual.

3.4.4.2.6 Operating instructions on decals and instruction plates. The location of decals and instruction plates affixed to the equipment shall be illustrated. Operating instructions in this section shall be written utilizing the information on the decals and instruction plates to the maximum.

3.4.4.3 Section III - Operation under unusual conditions. This section shall contain instructions for obtaining optimum operation of the equipment under unusual conditions. In addition to special operating procedures for each condition, instructions shall be included for all protective and preventive measures to be taken both at the time of operation and on a periodic basis. These preventive and protective measures shall be in addition to, but not part of, the operator's normal periodic maintenance services. Preventive or protective measures beyond the operator's capabilities shall be referenced to higher level maintenance facilities. The instructions shall include special procedures required for operation under the following conditions, as applicable:

3.4.4.3.1 Operation in unusual weather. The instructions shall include special procedures required for operation under conditions of extreme moist heat, extreme dry heat, extreme cold, salt air, sea spray, duststorms, sandstorm, high altitudes, sand, snow, mud, or similar conditions.

3.4.4.3.2 Fording. The instructions shall include special procedures required before, during and after fording the equipment.

3.4.4.3.3 Emergency procedures. Instructions shall be included for temporarily adapting the equipment and the operating procedure to meet reduction of power, partial failure, failure of a portion of the equipment, or similar conditions when continued equipment use is required.

3.4.4.3.4 Jamming and electronic countermeasure procedures. Instructions shall be included in this section for operation of the equipment in an environment of electronic countermeasures (man made interference, etc.). These instructions shall include procedures for operating the equipment through transmitted and reflected deception signals and transmitted and reflected jamming.

3.4.4.3.5 Demolition to prevent enemy use. Instructions shall be included in this section for destruction of materiel, armament equipment, communication equipment, and any auxiliary equipment if applicable.

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3.4.5 Maintenance instructions chapter. The objective of this chapter is to accurately present to the operator all the instructions and additional information needed by the operator to help him keep his equipment in good running order. This chapter shall consist of the following sections:

Section I	- Tools and equipment
Section II	- Preventive maintenance checks and services
Section III	- Lubrication instructions
Section IV	- Troubleshooting procedures
Section V	- Maintenance procedures

3.4.5.1 Section I - Tools and equipment. This section shall contain any additional tools, special test equipment, or supplies authorized by table of equipment which are required for operator's maintenance but are not listed in the Basic Issue Item List and Items Troop Installed or Authorized List and a statement similar to the following:

> Repair parts, special tools, test equipment and accessories issued with, or prescribed for use by, the operator for the (insert end item nomenclature) are listed in the Basic Issue Items List and Items Troop Installed or Authorized List, Appendix (insert applicable letter) of this manual.

3.4.5.2 Section II - Preventive maintenance checks and services (PMCS). This section shall contain the following.

3.4.5.2.1 General. The opening paragraph in the PMCS section shall contain a statement similar to the following:

> Be sure your (insert end item nomenclature) is always ready for operation. Inspect it systematically to discover and correct defects before they result in serious damage or failure.
>
> Necessary preventive maintenance checks and services to be performed are listed and described in figures (insert applicable figure numbers).
>
> Note defects discovered during the operation of the unit for future correction to be made as soon as operation has ceased. Stop operation immediately if a deficiency is noted during operation which would damage the equipment.
>
> Record all deficiencies together with the corrective action taken on (insert applicable form number).

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3.4.5.2.2 Equipment removal for PMCS. When the removal of an assembly or equipment, or the specific checks and adjustments to be made, requires the services of a higher level of maintenance, the PMCS shall be preceded by the following statement:

> Remove (insert name of equipment) from (insert name of equipment) every (insert proper interval) and have it checked by (insert authorized maintenance personnel).
>
> Instructions for performing the required checks are identified as periodic checks in the (insert pertinent maintenance level) maintenance manual.
>
> When the equipment has lubrication requirements, the following statement shall be entered on the tabular listing immediately preceding the procedures:
>
> > Lubricate in accordance with (insert reference to pertinent lubrication order for instruction in the technical manual, as applicable).

3.4.5.2.3 Special instructions. For equipment which is likely to be kept in continuous operation, the following note shall be included between the chart title and the chart:

◀ NOTE ▶

> If the equipment must be kept in continuous operation, check and service only those items which can be checked and serviced without disturbing operation.
>
> Make the complete checks and services when the equipment can be shut down.

If an item in a longer interval chart requires more frequent checking and servicing when the equipment is used in unusual environment (see 3.4.4.2) the special intervals shall be indicated by an asterisk or similar indicator before the sequence number, and a letter after the sequence number. Footnotes shall be used to explain special intervals.

3.4.5.2.4 PMCS procedures. This coverage shall provide the minimum requirements for operator's preventive maintenance checks and services which will insure that the equipment is operable and capable of performing it's intended mission. The following guidelines shall be used:

(a) The services will be arranged in a walk-around (as applicable) technique and in a prescribed sequence that will result in the minimum movement on the part of the operator. Arrange steps to prevent duplication of effort; such as, opening of panels, removal of access plates, etc.

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(b) Sequence numbers will be used as follows:

1. For "Before," "During" and "After Operation" intervals, sequence numbers will be in numberical order beginning with number 1 in the left "Before Operation" interval column for the first logical check until all applicable checks for that interval column are numbered. Sequence numbers shall continue consecutively throughout the remaining interval columns in logical order of performance.

2. For all other intervals (i.e., daily, weekly, monthly, etc.), each interval column will be numbered consecutively beginning with number 1.

(c) Checks should specify under what conditions they are made, e.g., engine warm, cold, etc.

(d) The PMCS shall include complete information required to accomplish the check or service, including supporting illustrations. Minimum references shall be made. The PMCS chart may include a style presentation applicable to both the experienced and inexperienced operator and shall state the following:

1. What is to be checked.

2. How often it is to be checked.

3. Minimum performance standards.

4. Any special inspections required under special conditions.

5. The time (in minutes) required to perform each check or service and a total time for each phase or section.

A typical example of a preventive maintenance checks and services chart is shown in Figure 27.

3.4.5.3 Section III - Lubrication instructions. This section shall contain or reference pertinent lubrication orders. Whether the lubrication orders will be contained in the manual, or referenced, will be specified by the procuring activity (see 6.2). Pertinent lubrication instructions not contained in a referenced lubrication order shall be provided here. For instructions on how to prepare formal lubrication orders or charts, see MIL-L-22690. If no lubrication is required, a statement to that effect shall be included. A typical example of formal lubrication instructions required for complex equipment is shown in Figure 28. An example of a less formal lubrication instructions that is acceptable for for simple equipment is shown in Figure 29. This section shall begin with an alphabetical index of the section contents if reference to this section is included in front cover table of contents (see 3.4.1.3).

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Table 2-1. Preventive-Maintenance Checks and Services - Continued

B - Before operation D - During operation A - After operation

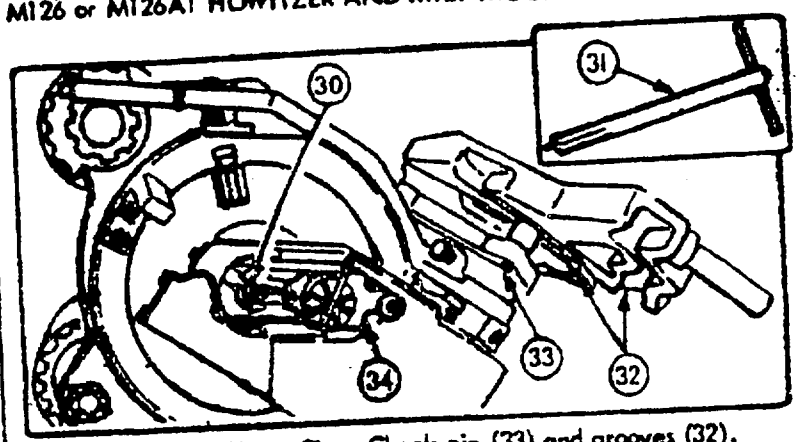
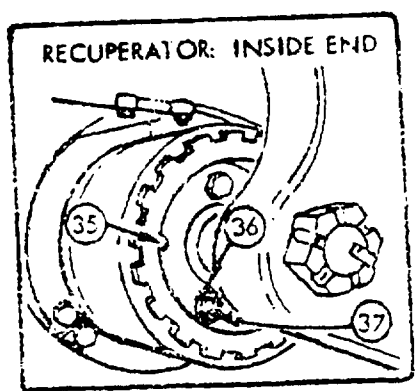
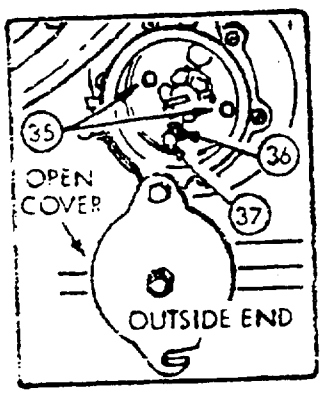
Interval and sequence no. B	D	A	ITEM TO BE INSPECTED Procedure	WORK TIME (MINUTES)
			M126 or M126A1 HOWITZER AND M127 MOUNT - Continued	
				
39			Cradle Cam Hinge Pin. Check pin (33) and grooves (32). Lube grooves daily when cannon is being fired.	6
40		65	Firing Mechanism. Check operation of firing mechanism (34) with a steady pull on the lanyard before operation. Wipe clean and coat with oil (PL-S) after operation. Place in firing position. Check primer chamber (30) for rust or powder fouling. Clean chamber by inserting reamer (31) as far as it will go, and rotate briskly.	6
				
				
41			Recuperator. Check pins (35) before firing. Pins must not extend over 3/4 inch (1/8 inch is correct). If pins (35) do extend beyond 3/4 inch, remove cap (37) from valve (36) and add hydraulic fluid (OHT) until pins extend 1/8 inch. Note: The recuperator is charged with a pressure of 550 PSI in the M109 and 700 PSI in the M109A1. With temperature at 100° or over, 600 PSI in the M109A1 is recommended.	6

Figure 27. Preventive Maintenance Chart Format

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LUBRICATION INSTRUCTION **LI-02550C**

TRUCK, LIFT, FORK, ROUGH TERRAIN

OS-3354

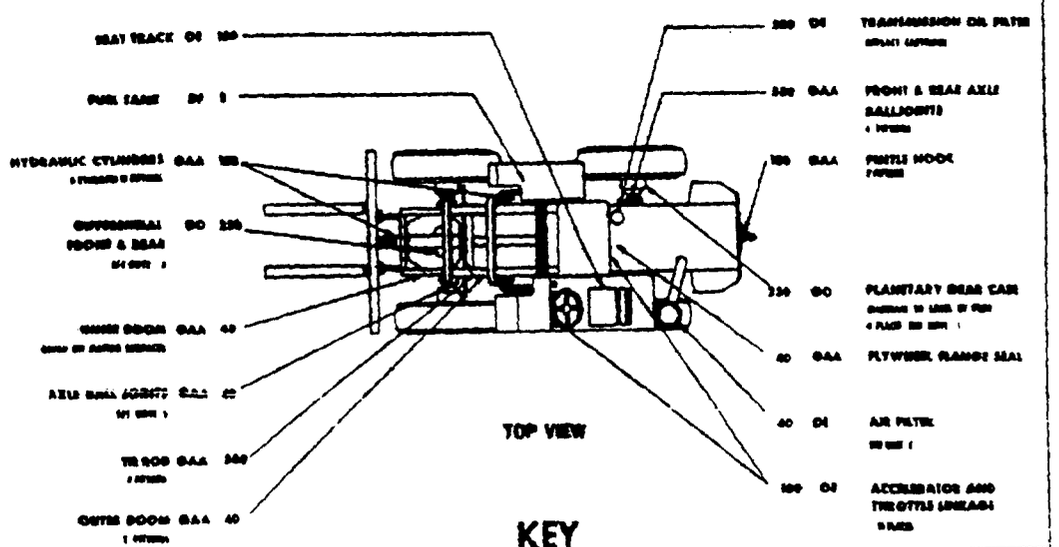
Copy of this instruction to be carried with the equipment at all times. Clean fittings before lubricating. Lubricate after washing equipment. Inspect and lubricate more frequently under severe operating conditions. Relubricate as soon as practicable after exposure to salt water.

Equipment in storage or received direct from the manufacturer is normally preserved with special preservatives and lubricants. On receipt of equipment, service all components in accordance with this instruction prior to operation.

See reverse side of instruction for additional points requiring lubrication.

LUBRICANT @ INTERVAL

INTERVAL @ LUBRICANT



KEY

SYMBOL	LUBRICANTS AND APPLICATIONS	CAPACITY	EXPECTED TEMPERATURES			INTERVALS	
			ABOVE +32°F	+32°F TO 0°F	0°F TO -65°F	SYMBOL	INTERVAL
OE	OIL, ENGINE ENGINE CRANKCASE, AIR FILTER	15 QTS 3 QTS	MIL-L-2104 OE-30	MIL-L-2104 OE-10	MIL-L-10295 OES	8 40 100 250 500	8 HRS 40 HRS 100 HRS 250 HRS 500 HRS
	HYDRAULIC SYSTEM TORQUE CONVERTER	42 GAL 18 QTS	MIL-L-2104, OE-10				
GAA	GREASE, AUTOMOTIVE & ARTILLERY		MIL-G-10924				
GC	OIL, LUBRICATING, GEAR STEERING GEAR		MIL-L-2105				
HB	HYDRAULIC FLUID BRAKE		VV-B-680 HB		MIL-H-13910 HBA		
DF	DIESEL FUEL FUEL TANK	62 GAL	VV-F-800 DF-2	VV-F-800 DF-1	VV-F-800 DF-A		
COOLANT	COOLING SYSTEM	26 QTS	WATER & RUST INHIBITOR	50% WATER & 50% ETHYLENE GLYCOL	ANTIFREEZE ARCTIC MIL-C-11755		

Figure 28. Lubrication Instruction

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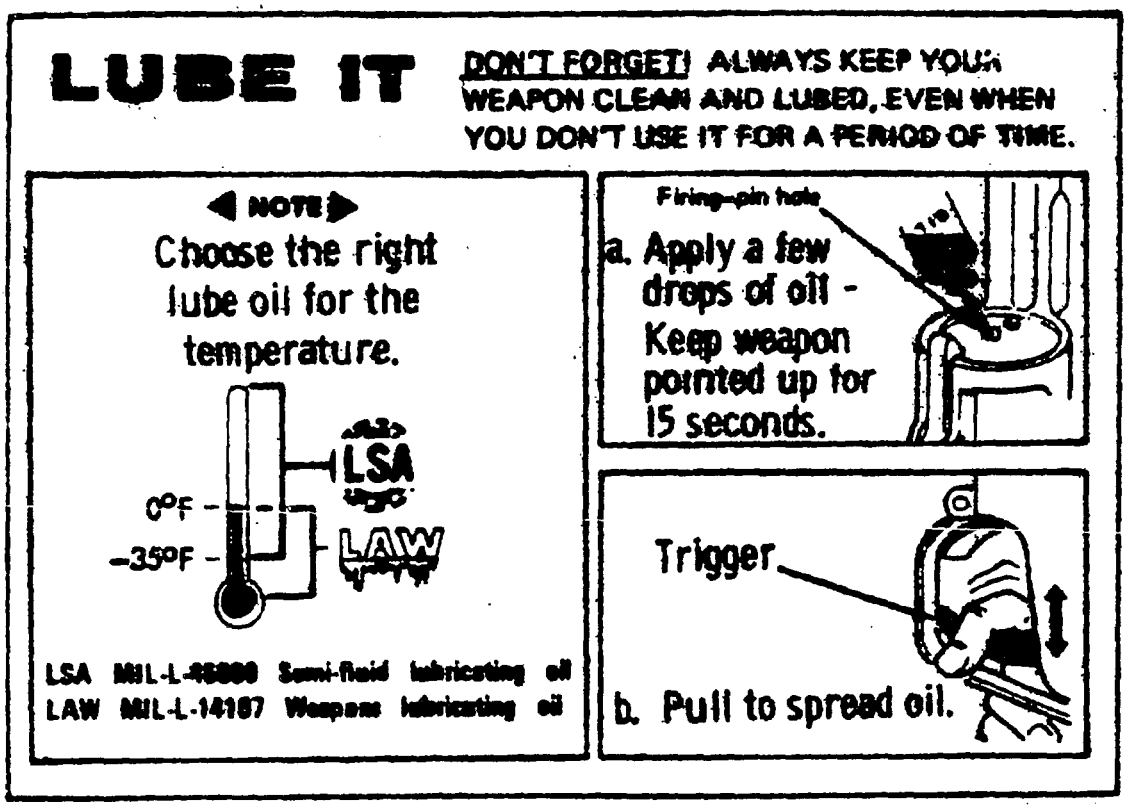


Figure 29. Informal Lubrication Order

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3.4.5.4 Section IV - Troubleshooting procedures. This section shall contain instructions to allow the operator to efficiently troubleshoot the equipment. There may also be a symptom index located in the front of the section.

3.4.5.4.1 Troubleshooting procedure guidelines. The troubleshooting procedures shall be based on the following guidelines:

(a) The instructions shall be based upon symptoms which may be observed by the operator during preventive maintenance checks and services as well as during normal operation of the equipment.

(b) Only those checks and corrective actions which are specified by the procuring activity (see 6.2) shall be included.

(c) Tests and inspections will be listed in logical sequence for each malfunction listed. The test and inspection will be complete so as to insure that the malfunction can be isolated and verified. In determining logical sequence, consideration should be given to:

1. Location and accessibility of components, subassemblies, and assemblies.

2. Time required to perform the test or inspection

3. Probability of being the cause of the malfunction.

(d) Corrective action procedures shall include minimum reference to information in Section III - Lubrication instructions and Section V - Maintenance procedures, as required. If the required information does not exceed 120 words or 10 lines, then it should be duplicated in the troubleshooting chart rather than referenced.

A typical example of a troubleshooting procedure is shown in Figure 30. For electronic equipment, the troubleshooting concept and format may be designed to fit the particular application.

3.4.5.4.2 Symptom index. There may be a symptom index located in the front of the troubleshooting section to assist the operator in locating the correct troubleshooting procedure. The entries shall be symptoms which are observable by the operator during normal operation of the equipment and during preventive maintenance checks and services. A typical example of a symptom index is shown in Figure 31. Also, see 3.4.11.2 for additional description of this type of index.

3.4.5.5 Section V - Maintenance procedures. This section shall contain maintenance procedures which are the responsibility of the operator. The section may contain each of the following topics, as applicable:

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Table 3-1. Troubleshooting

MALFUNCTION TEST OR INSPECTION	CORRECTIVE ACTION
Note : For corrective actions of malfunctions not listed in this table, notify your organizational maintenance personnel.	
Note : Wherever the word lubricate appears, see LI-09346A.	
ENGINE	
1. ENGINE FAILS TO CRANK, OR CRANKS SLOWLY, WHEN STARTER SWITCH IS ACTIVATED.	
Step 1. Check to see if MASTER SWITCH is on.	Turn on.
Step 2. Check to see if transmission lever is in N position.	Place to N position.
Step 3. Check to see if battery cables are loose, broken, or corroded.	Clean and tighten loose connections at battery, ground, and starter. If cables are broken, notify organizational maintenance.
Step 4. Check to see if your battery indicator reads charge/discharge.	Push or use auxiliary power to start engine; then if it doesn't start, notify organizational maintenance.
Step 5. Check to see if you have correct weight oil for weather conditions.	Drain and refill with correct weight (LI-09346A).
2. ENGINE CRANKS BUT FAILS TO START.	
Step 1. Check to see if fuel gage indicates empty.	Fill, if no fuel.
Step 2. Check to see if fuel SHUT-OFF control knob is pulled out.	Push in knob.
Step 3. Check for restricted fuel lines and hoses.	Disconnect and straighten kinked/pinched tubes or hoses.

Figure 30. Troubleshooting Procedure Format

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SYMPTOM INDEX

	Troubleshooting Procedure Page
COOLING SYSTEM	
Radiator	
Boils over	3-12
Leaks	3-17
Temperature Gage	
No indication	3-29
Runs cold	3-11
Runs hot	3-13
ENGINE	
Misses	3-34
Overheats	3-72
Won't start	3-27
EXHAUST SYSTEM	
Excessive smoke	3-33
Water vapor	3-33

Figure 31. Symptom Index

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(a) Introduction.
(b) Inspection.
(c) Adjustment/Alignment. (Include instructions for servicing or adjusting without removing mechanism.)
(d) Checks and Adjustments (Maintenance calibration).
(e) Removal and Installation.
(f) Disassembly and Assembly.
(g) Repair.
(h) Cleaning.
(i) Test After Repair.

Examples of maintenance procedures are shown in Figures 32 and 33. Additional methods may be devised as appropriate to portray the best repair instructions possible for the specific equipment covered by the manual. This section shall begin with an alphabetical index of the section contents if reference to this section is included in the front cover table of contents (see 3.4.1.3).

3.4.6 Maintenance of auxiliary equipment chapter. If there is no auxiliary equipment, this chapter may be omitted. Maintenance procedures which are the responsibility of the operator shall be described for all accessories and attachments which are used in conjunction with the major item. For auxiliary equipment procured separately from the basic equipment or when the basic equipment will be used with existing equipment, the contractor shall obtain data on the auxiliary items from the procuring activity and shall provide all information required for maintaining the basic and auxiliary equipment as a unit. The topics covered in this chapter are the same as those listed in 3.4.5.5. If maintenance of any item is covered in a separate manual, reference shall be made to that manual unless the material is less than two pages or can be condensed to two pages. This chapter shall begin with an alphabetical index of the chapter contents if reference to this chapter is included in the front cover table of contents (see 3.4.1.3).

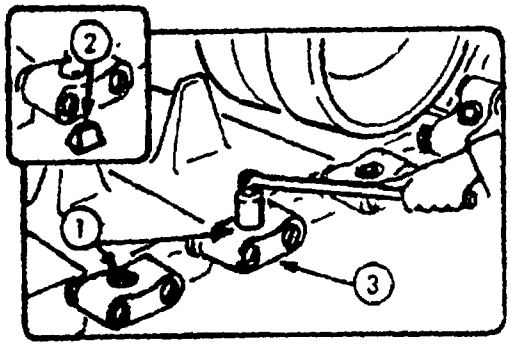
3.4.7 Ammunition chapter. This chapter shall be included in all operator's manuals on weapons involving the use of ammunition. It shall contain only that information prescribed for the weapon and needed by the operator. An illustration shall be used to clearly show all identifying markings. See Figure 34 for a typical example. A table shall be used covering classification, identification, care, handling, preservation, transportation, authorized rounds, preparation for firing, fuzes, packing, and other pertinent data and references, as appropriate. See Figure 35 for a typical example.

3.4.8 Appendix A - References. Information contained in this appendix shall be furnished by the procuring activity when specified (see 6.2). (See Figure 36 for a typical example.) See the topic, "References," in MIL-M-0038784 for general requirements.

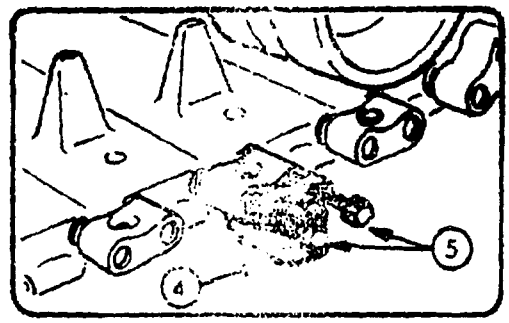
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REMOVING/INSTALLING TRACK SHOE

1 Decrease track tension.

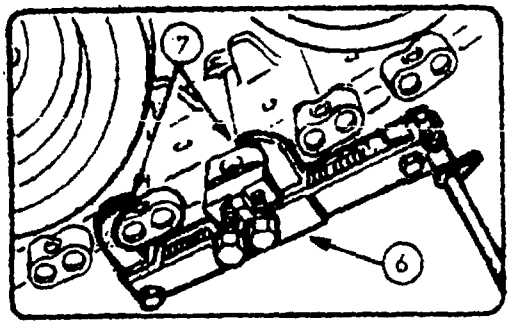


2 Remove bolts (1) retaining wedges (2) on all four end connectors (3) holding track shoes

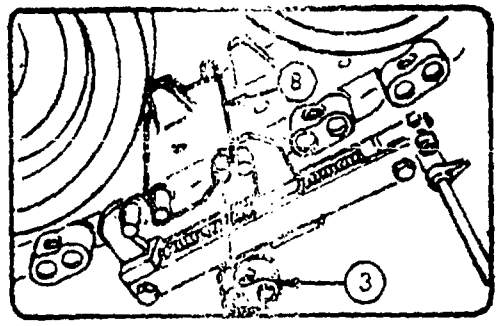


3 Install end connector puller (4) and tighten screws (5) against ends of track link pins.

4 Move all four end connectors out approximately 1 inch, but do not remove.



5 Install track fixtures (6), on both sides of shoe being removed, engaging track link pins (7) on adjoining shoes.



6 Remove end connectors (3) and track shoe (8).

7 Install new track shoe by reversing procedures 6, 5, 4, 2, and 1.

8 Tighten wedge screws to 75-86 ft lbs if torque wrench is available.

DISCONNECTING TRACK

1 Disconnect track in same manner as replacing track shoe, but only remove two connectors opposite each other.

2 Before removing track fixtures, support weight of track with crow bar to lower track to ground.

WARNING: Keep your toes clear of track.

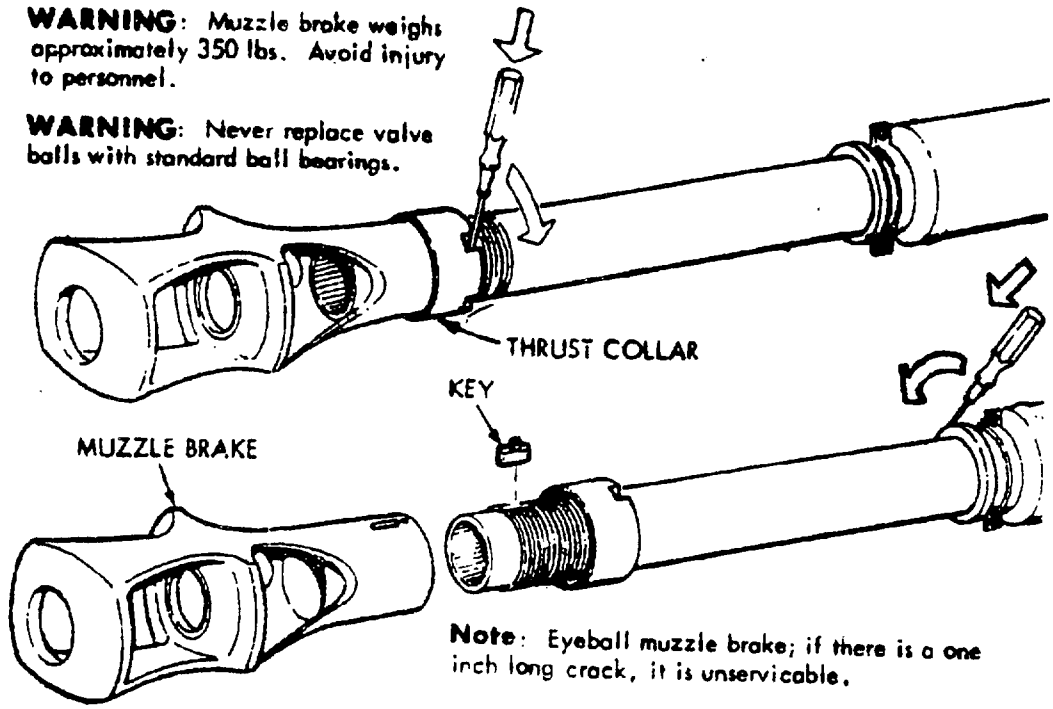
Figure 32. Maintenance Procedure Format (Example 1)

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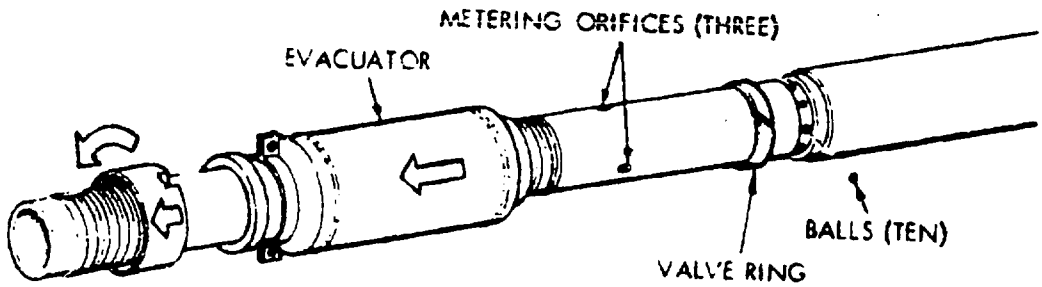
REMOVING/INSTALLING M109A1 MUZZLE BRAKE AND EVACUATOR

WARNING: Muzzle brake weighs approximately 350 lbs. Avoid injury to personnel.

WARNING: Never replace valve balls with standard ball bearings.



Note: Eyeball muzzle brake; if there is a one inch long crack, it is unservicable.



1 Depress thrust collar detent with screwdriver and screw collar rearward enough to uncover key.

2 Remove key from brake and tube.

3 Unscrew muzzle brake until threads in muzzle brake just clear threads on cannon tube. Attach a sling and hoist to muzzle brake and remove brake from tube.

4 Screw thrust collar forward and remove.

5 Depress evacuator detent, unscrew evacuator and slide forward.

6 Slide valve ring forward and catch balls.

Note: Valve seats (10) and metering orifices (3) may be cleaned with muzzle brake still in position.
(LI-08384B)

Figure 33. Maintenance Procedure Format (Example 2)

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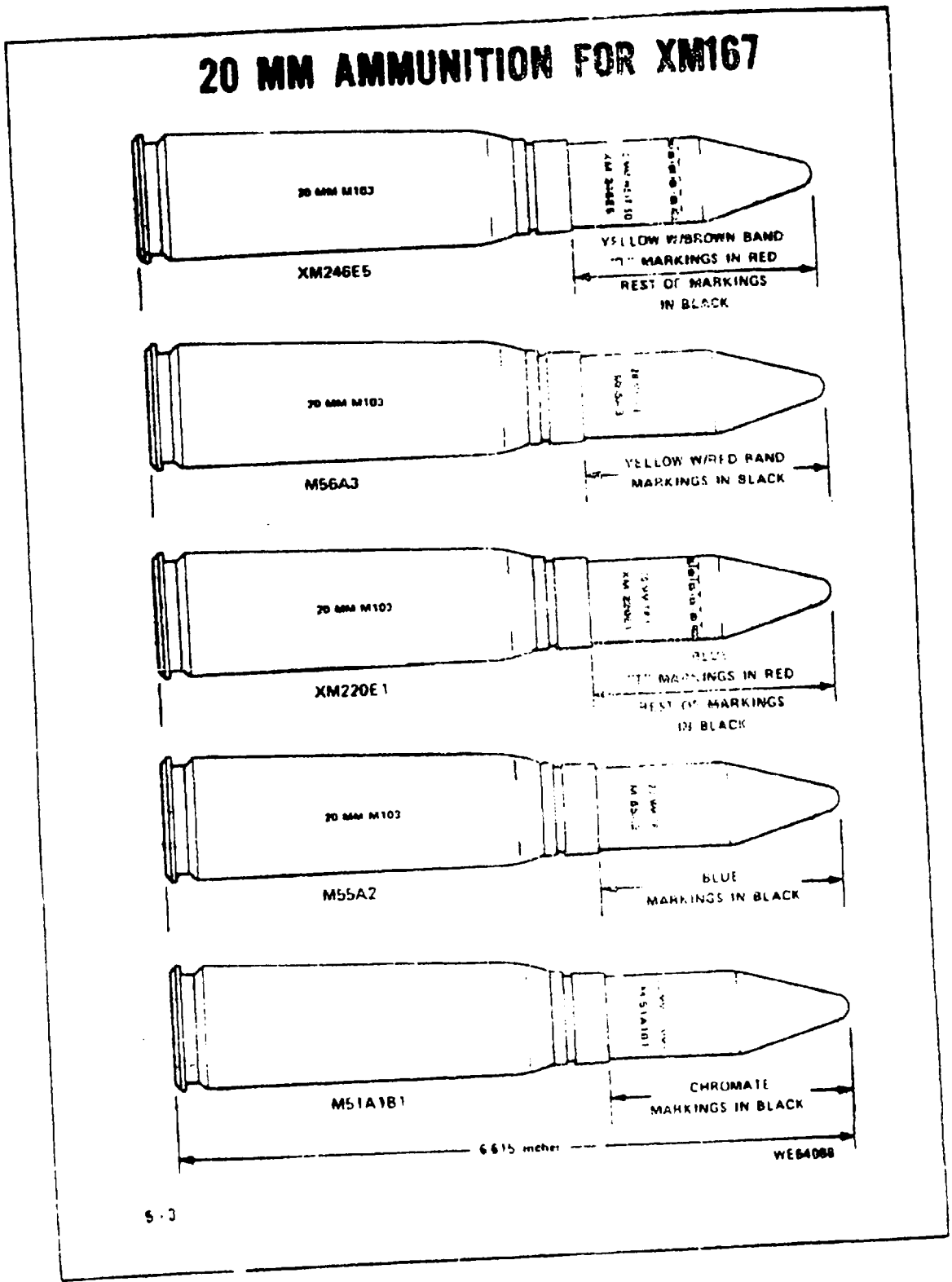


Figure 34. Ammunition Identification Illustration

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20 MM AMMUNITION TABLE FOR XM167

Care:	Keep ammunition clean of dirt and grease. Do not use deformed ammunition.
Handling:	Avoid excessive shock, electrical discharge, or excess temperature.
Preservation:	Do not open sealed ammunition containers until just before use.
Packing:	Pack unused shells in original containers and use first in subsequent firings.

Authorized Rounds:	Classification:	Identification:	Fuses:	Remarks:
HEIT-SD, XM246E5	High-explosive incendiary tracer self-destruct	• Yellow w/brown band projectile • "T" in red on black markings on case	P.D.	For aerial targets
HEI, M56A3	High-explosive incendiary	• Yellow w/red band projectile • Black markings on case	P.D.	For ground targets
TP-T, XM220E1	Target practice tracer	• Blue projectile • "T" in red on black markings on case	Dummy	
TP, M55A2	Target practice	• Blue projectile • Black markings on case	Dummy	
Dummy, M51A1B1	Dummy	• Chromate projectile • Black markings on case		Completely inert

5-2

Figure 35. Ammunition Information Table

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APPENDIX A

REFERENCES

The following publications should be consulted frequently for the latest changes or revisions, and for new publications relating to materiel covered in this manual.

A-1. INDEXES

DA PAM 310-3 ---------- Index of Training Publications
SL-1-2 ---------- Index of Authorized Publications for Equipment Support

A-2. SUPPLY CATALOGS

DASM 9-1 Series ---------- Stock List of End Items
DASM 9-5 Series ---------- Stock List of Current Issue Items
SL-4-07007A ---------- Marine Corps Stock List for LVTP7

A-3. OTHER PUBLICATIONS

LI-07007A/07268A-10 ---------- Marine Corps Lubrication Instruction for Landing Vehicle, Tracked, Personnel, Model 7 (LVTP7) and Landing Vehicle, Tracked, Command, Model 7 (LVTC7)

TM 3-4240-236-12 ---------- Operation and Organizational Maintenance Manual: Filter Unit, Gas Particulate, Tank, Four Man, 12 CFM, ABC-M8A3 w/chg 1

TM 4700-15/1A ---------- Marine Corps Tactical Equipment Record Procedure: Record Maintenance

TM 9-1300-206 ---------- Ammunition; Care; Handling; Preservation; and Distribution

TM 9-207 ---------- Operation and Maintenance of Army Materiel in Extreme Cold Weather, 0° to -65°F

TM 11-5820-401-10 ---------- Operator's Manual, Radio Sets AN/VRC-12 and AN/VRC -43, -44, -45, -46, -47, -48, and -49 (w/chg 1 & 4)

TM 11-5855-202-13 ---------- Operator, Organizational and DS Maintenance Manual (including repair parts and special tool list): Night Vision Sight, Crew Served Weapons, AN/TVS-2 (w/chg 1)

TM 21-301 ---------- Driver Selection, Training, and Supervision; Tracked Vehicles

TM 21-306 ---------- Manual for the Full Tracked Vehicle Driver

Figure 36. References Appendix

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Check the source to verify that this is the current version before use.

3.4.9 Appendix B - Basic issue items list and items troop installed or authorized list. The basic issue items list and items troop installed or authorized list shall be prepared in accordance with MIL-M-63001 or an informal method may be used for illustrating simple equipment as shown in Figure 37.

3.4.10 Glossary. If the manual contains a number of terms requiring definition, a glossary shall be included (see Figure 13 for typical example).

3.4.11 Indexes. Indexes shall be prepared as follows.

3.4.11.1 Alphabetical indexes. The alphabetical subject index shall be prepared in accordance with MIL-M-0038784. Headings within the index shall include controls, procedures, indicators, common names used by the operator, systems and subsystems, and other descriptions (see Figure 38 for a typical example).

3.4.11.2 Other indexes. Other indexes shall be included as determined necessary to improve user access to required information. These may be by location, configuration, serial number, subassembly, auxiliary test equipment, or any outstanding similarity or difference which will help the user find the information he needs in the manual. Examples of sections which may have their own index at the beginning of the section are the following:

(a) Operating Instructions
(b) Preventive Maintenance
(c) Repair Instructions
(d) Checks and Adjustments
(e) Troubleshooting Procedures

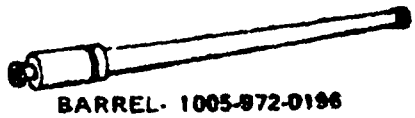
These indexes will normally be used in conjunction with the cover table of contents. For example, the cover table of contents might state that operating instructions begin on page 2-1; the index on this page helps locate the exact procedure wanted.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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BASIC ISSUE ITEMS must be with your MACHINEGUN:

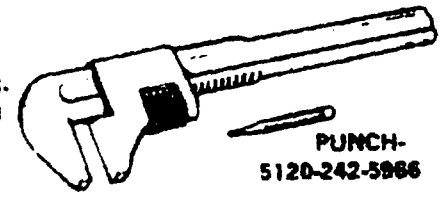


BARREL- 1005-972-0196

COMBINATION TOOL- 4933-733-4759



WRENCH- 5120-264-3793



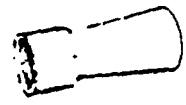
PUNCH- 5120-242-5966



RUPTURED CARTRIDGE CASE EXTRACTOR- 4933-652-9950

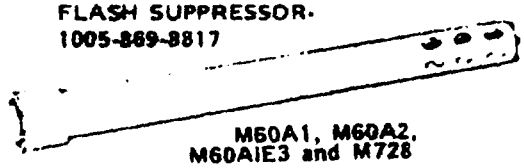
BASIC ISSUE ITEMS must be with your VEHICLE:

FLASH HIDER- 1005-922-9777



M48A3, M60, M551 and XM706

FLASH SUPPRESSOR- 1005-869-8817



M60A1, M60A2, M60AIE3 and M728

BARREL JACKET EXTENSION- 1005-933-8070



M60A2 ONLY

ITEMS TROOP INSTALLED OR AUTHORIZED

These items, at the discretion of your CO, may accompany your machinegun, but are not required to be turned-in with the machinegun.

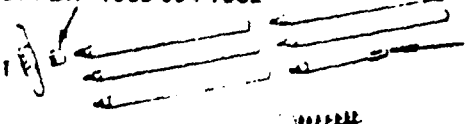
SMALL ARMS CLEANING ROD CASE- 1005-550-6573



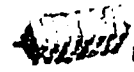
BUFFER- 1005-694-1662

CLEANING ROD SECTION (5)- 1005-726-6109

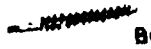
HANDLE- 1005-793-6761



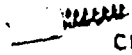
SWAB HOLDER SECTION-1005-726-6110



RECEIVER BRUSH- 1005-650-4508



BORE BRUSH- 1005-556-4174



CHAMBER BRUSH-1005-690-3115

CONSUMABLE EXPENDABLE ITEMS

PL Special	4 oz.	9159-273-2389	SD	1 gal.	6850-281-1985
LAW	1 qt.	9150-292-9689	RBC-	6 oz.	6850-224-6657
Swabs	1,000	1005-288-3565	LSA-	4 oz.	9150-889-3522

Figure 37. Informal Sample of Basic Items Troop Installed

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ALPHABETICAL INDEX

Subject	Page Number
A	
Accessory Section	
Installation	8-24
Removal	4-54
Actuating Pilot Valve Leakage Check	
Actuator, Compressor Bleed Valve	
Assembly	10-111
Cleaning	10-108
Disassembly	10-107
Inspection	10-109
Repair	10-110
Adapter, Compressor Repair	6-135
Afterburner	
Description	2-29
Installation	8-110
Performance Limits	
Removal	4-141
Air System	2-74
Airseal Installation	8-9
Anti-icing Air System Description	2-17
B	
Baffle Assembly	6-28
Baffle and Spacer Inspection	6-29
Bearings, Anti-friction	
Balance	6-15
Cleaning	6-16
Bleed Control Limit Curve	
Breather Pressurizing Valve	
Disassembly	2-22
Installation	5-105
C	
Carbon Seals	
Cleaning	3-14
Inspection	3-60
Combustion Chambers	
Igniters	3-128
Inner Liners	3-127
Repair	3-126
Compressor Bleed Valves	
Check	6-39
Ground Safety Precautions	6-7

Figure 38. Index Format

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4.2 Contractor inspection. Material furnished in accordance with this specification shall be inspected by the supplier for conformance to the applicable requirements herein.

4.3 Government inspection. Material furnished in accordance with this specification shall be subject to inspection, verification, and approval or disapproval by the Government as specified by the terms of the contract. Inspection verification will be performed by the Government prior to acceptance.

4.4 In-Process reviews. When specified by the procuring activity (see 6.2), these reviews shall be performed by its representatives during preparation of the manual, primarily as guidance to the contractor and to assure that the manuals are written in conformance with contract and specification requirements. The reviews may be conducted at the contractor's facility, or at the procuring activity's facility, at any time during the development of the manual but normally prior to preparation of the final draft. These in-process reviews are not a part of validation or verification and shall not be used to take the place of validation verification.

4.5 Validation. The draft manual shall be validated by the contractor in accordance with the validation schedule. A 100-percent validation of the operating and maintenance procedures shall be performed for technica accuracy and adequacy of content. Validation of readability shall conform to the requirements in 4.5.6. Validation shall consist of the following:

(a) Validation includes, but is not limited to, the actual performance by contractor personnel of operating and maintenance procedures. It includes review of instructions for checkout, calibration, alignment, scheduled removal and replacement, and a review of associated checklists to assure technical accuracy and adequacy.

(b) All disassembly, cleaning, inspection, testing, repair, replacement, reassembly, troubleshooting, preventive maintenance checks and services, and similar maintenance procedures shall be validated by actual performance. All other data, such as schematic diagrams, wiring data, and descriptive data, shall be checked to assure accuracy.

4.5.1 Validation schedule. Within the number of days specified by the statement of work portion of the contract, the contractor shall provide the technical office identified in Item 6 of DD Form 1423, Contract Data Requirements List, with a schedule for accomplishment of validation requirements. The procuring activity reserves the right to inspect, reject, and require revision of the validation schedule. This right shall be exercised with.n the number of days specified by the statement of work portion of the contract. Otherwise, it will be assumed that the validation schedule is acceptable as written. Reviews may be performed on a chapter-by-chapter basis during preparation or upon completion

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of the draft manual. The validation schedule shall indicate (1) the calendar dates for performance of the validation; (2) the location where validation is to be conducted; (3) whether the validation will be on the equipment or simulated; and (4) whether the validation will be on a chapter-by-chapter basis or on the complete manuscript.

4.5.2 Performance. The draft manual shall be reviewed for:

(a) Conformance to applicable requirements of the governing documents. This review shall include editorial review of the manuscript.

(b) Technical accuracy and adequacy of the content. This review shall include the actual performance or simulation of the operating and maintenance instructions on the equipment system for which the draft manual was written. Simulation shall be used only when the procuring activity has approved this method by its acceptance of the validation schedule. This validation shall include but not be limited to a review for:

(1) Maintenance instructions for all functions specified in the maintenance allocation chart.

(2) Repair parts, special tools, and test equipment required to perform functions specified in the maintenance allocation chart.

(3) Essentiality of illustrations cited in the text.

(4) Existence and adequacy of references cited in the text.

(5) Instructions for manufacturing items source coded "M", for assembling items source coded "A", and for repairing items coded as recoverable, reparable.

4.5.2.1 Revalidation. All material changed or corrected as a result of engineer tests, service tests, user coordination, and validation verification shall be revalidated to meet the validation requirements of this specification.

4.5.2.2 Separation of responsibility. The Government reserves the right to require that contractor personnel performing operating and maintenance instructions on the equipment during validation be independent from the contractor's manuscript preparation activity.

4.5.3 Correction of errors. The contractor shall correct all contractor errors found in the manuscript during validation at no cost to the Government. After Government acceptance of the manuscript, the Government reserves the right to require the contractor to correct all errors found in the manuscript, at no cost to the Government, within the number of days specified in the statement of work portion of the contract.

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The contractor is not liable for any equipment adversely affected by errors in the manuscript.

4.5.4 Records. The contractor shall maintain records showing dates of validation reviews, manuscript material reviewed, findings, and action taken. The procuring activity reserves the right to examine these records at the contractor's facility.

4.5.5 Government-provided information. The contractor is not responsible for the adequacy and accuracy of Government furnished information provided for inclusion in the manuscript. The contractor shall include in his reviews a validation that all such Government-furnished information is properly and correctly reflected in the manuscript. The contractor shall be responsible for notifying the procuring activity of any inaccurate or inadequate data or of any data which is inconsistent with the contents of the manuscript.

4.5.6 Validation of readability. Narrative text shall be validated for conformance to the readability standards specified in 3.1. If one or more of the standards are exceeded, the manuscript shall be rewritten as required to meet the standards. Random sampling shall be used. Selected samples shall represent as equally as possible the various types of narrative material; such as operation, description, maintenance, and troubleshooting instructions. The number of samples taken, using 100-word per page units, shall be in accordance with the following:

From Total Pages Narrative	Number of Samples
Under 26	3 minimum
26 to 90	5
91 to 150	8
151 to 280	13
281 to 500	20
Over 500	32 maximum

4.5.6.1 Sample marking. The sample shall be marked as follows:

(a) The sample shall start at the beginning of a sentence. Headings and captions shall not be included in the sample.

(b) All words, including nomenclature, alphameric expressions, symbols and abbreviations which are surrounded by white space shall be counted in sequence to reach the 100-word total. The following are examples counted as one word (10-foot, 1.345, M14A1, G's, -10 degrees F, etc.).

4.5.6.2 Average senctence length (standard No. 1). The average sentence length shall be computed as follows:

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(a) Total all words in all samples. This figure will be in even hundreds.

(b) Total all sentences in all samples. If the 100-word mark occurs within a sentence but after half of the words in that sentence, count the sentence when totalling sentences. Otherwise, include the words necessary to achieve 100-word count but do not include the sentence in the sentence count.

(c) Divide the total number of words by the total number of sentences. Carry the quotient to hundredths and round it off to a whole number of words. (17.49 shall be rounded off to 17 and 17.50 shall be rounded off to 18.)

4.5.6.3 Average word length (standard No. 2). The average word length shall be computed as follows:

(a) Identify by underlining all mandatory words in the sample and count them.

(b) Add a corresponding number of nonmandatory words from the narrative that appears after the end of the sample and mark the new end point of the adjusted 100-word sample. The adjusted sample will now contain exactly 100 nonmandatory words.

(c) Count all syllables in the 100 nonmandatory words in the adjusted sample.

(d) Total all nonmandatory words in all samples. This figure will be in even hundreds.

(e) Divide total syllables by total words. Round off the quotient: e.g., round off 1.603 to 160 and 1.605 to 1.61.

4.5.6.4 Personal sentences (standard No. 3. The percentage of personal sentences shall be computed as follows:

(a) Take samples consisting of 100 sentences per sample using the sample pages already selected. Succeeding pages may be added to attain the 100 sentence requirement of each sample.

(b) Count the number of personal sentences in each sample.

(c) Total all sentences in all samples.

(d) Total all personal sentences in all samples.

(e) Divide total number of personal sentences by the total number of sentences. Convert quotient to percent.

4.6 Verification. Representatives of the procuring activity will verify the contractor's validation. In addition to reviewing the draft manual for conformance to the requirements of the governing documents, and inspecting the contractor's validation records, the Government reserves the right to perform verification by one or more of the following:

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(a) Witnessing the validation reviews performed by the contractor.

(b) Performing at the contractor's plant the contractor's validation review as scheduled by the contractor in accordance with the validation schedule.

(c) Reviewing the technical content of the draft manual by having Government personnel perform on the equipment, either at the contractor's plant or at Government facilities, the operating and maintenance instructions.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Packaging shall conform to requirements of method 1C of MIL-P-116 and the following.

5.1.1.1 Manuscript. Manuscript shall be packaged flat in cartons or wrapped separately. Elaborate containers are unnecessary. The illustration and text material may be separated in one package, or may be in separate packages. Large, unmounted illustrations may be rolled. Copies of large illustrations to be used for review purposes only may be folded. Packaging need only be adequate for providing for safe delivery and for providing protection against forms of damage that frequently occur during shipping.

5.1.1.2 Reproducible copy. Reproducible copy shall be packaged flat and double packaged. Illustrations shall not be folded or rolled. The interior material shall be waterproof and free of any chemical substance which would discolor or otherwise render the reproducible copy useless. The exterior package shall be a standard commercial carton at least equal to Interstate Commerce standards and of sufficient strength to provide for safety and safe delivery, and to protect the reproducible copy against forms of damage that frequently occur during shipping. Packaging that provides protection against seldom encountered extremes is unnecessary.

5.1.1.3 Classified material. Classified material shall be packaged in accordance with DOD 5200.1-R.

5.1.1.4 Original illustrations. When the procuring activity requires that the original illustrations be furnished, they shall be packaged as specified (see 6.2); for example, if an original oversize line drawing is on vellum or tracing cloth it could be rolled if the contract so authorizes.

5.1.2 Level C. Packaging shall be in accordance with the suppliers commercial practice and in a manner which will insure safe delivery at destination.

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5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The manuals prepared in accordance with this specification provide operational and maintenance instructions for equipment to the levels of maintenance indicated by the specific manuals concerned.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether a comprehensive outline is required (see 3.1).
(c) Size of manual (see 3.2).
(d) Whether separate lubrication orders exist (see 3.4.5.3).
(e) Checks and corrective actions required (see 3.4.5.4.1(b)).
(f) Whether the reference list (Appendix A) will be furnished (see 3.4.8).
(g) When in-process reviews are required (see 4.4).
(h) Level of packaging required (see 5.1).
(i) Packaging instructions for original illustrations (see 5.1.1.4).
(j) When marking shall not be as specified in 5.2.

Preparing activity:
Navy - MC
Project No. TMSS-N099

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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